

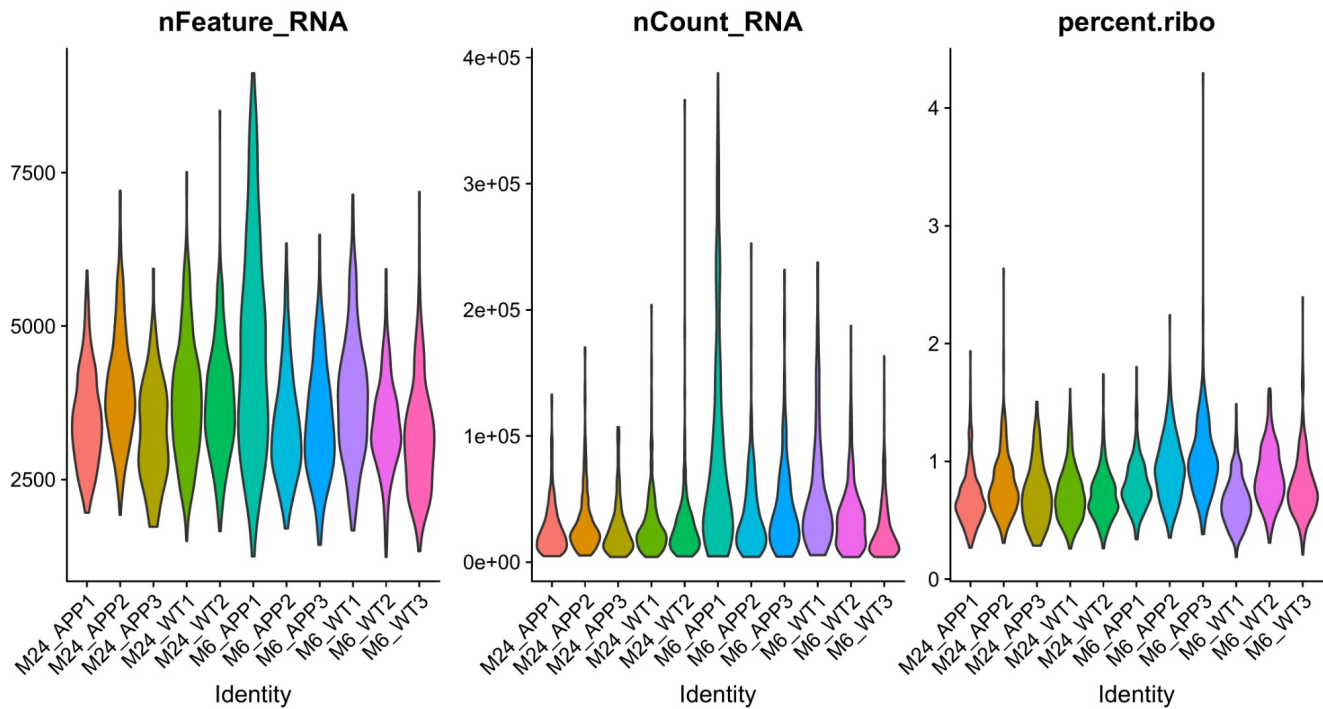
Supplementary Figures

APP23 hippocampal single-nucleus RNA-seq analysis

This file contains the supplementary analysis figures uploaded in the current project chat, excluding the figures already placed in the manuscript as main figures. Duplicate uploads were removed.

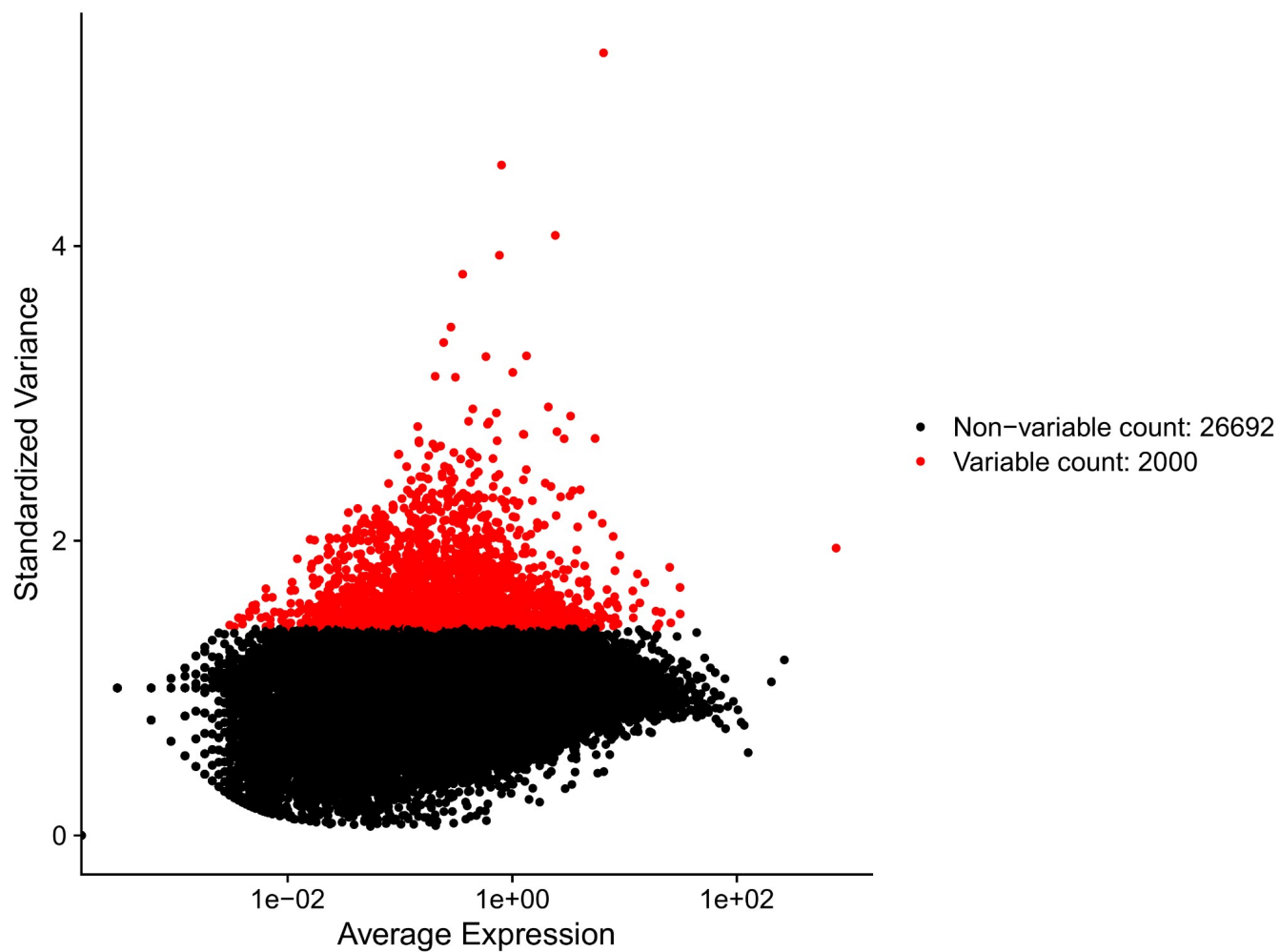
Supplementary Figure S1. Quality control, feature selection, and dimensionality reduction diagnostics.

A. Per-sample distributions of detected genes, transcript counts, and ribosomal percentage.

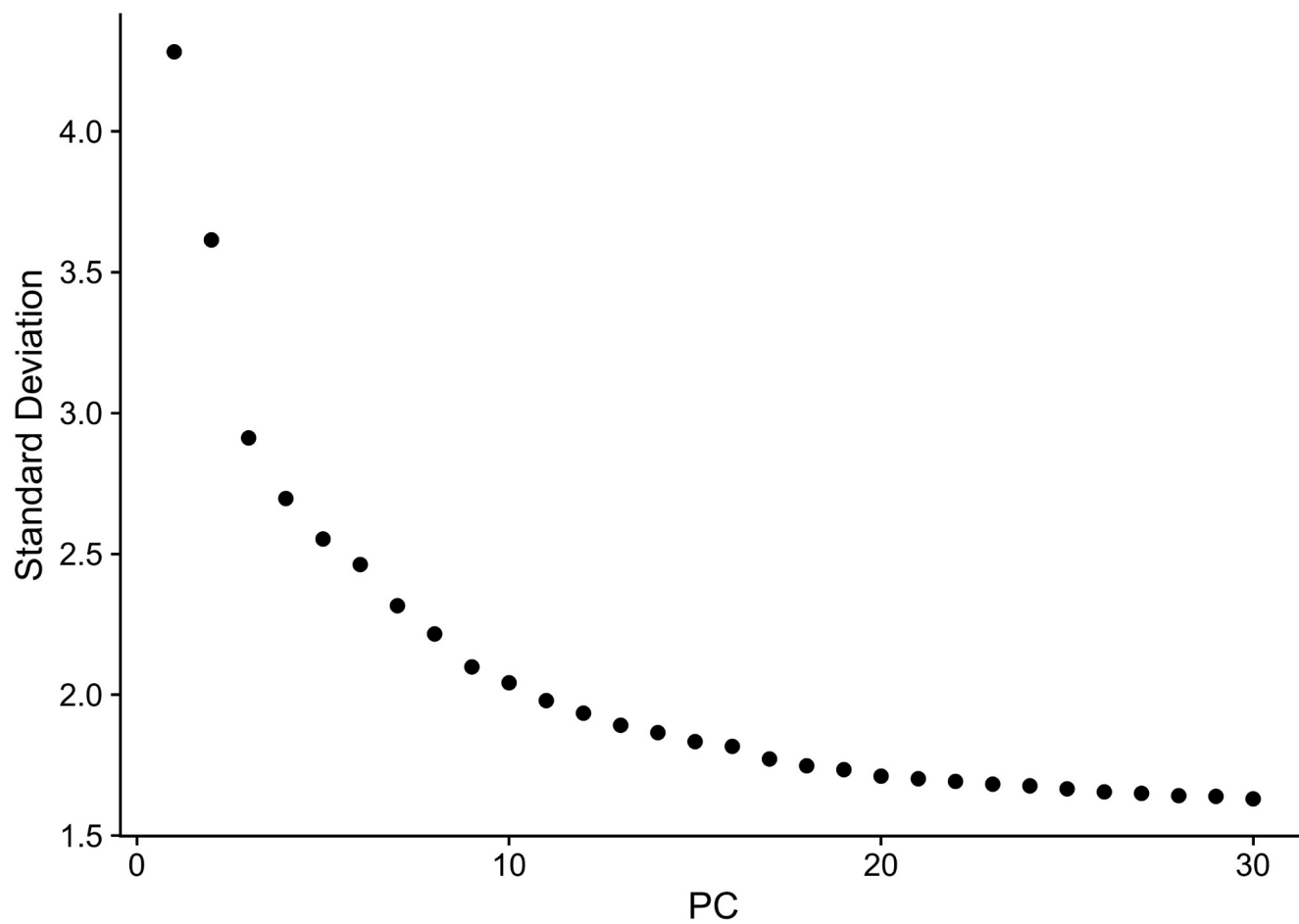


Supplementary Figure S1 (B)

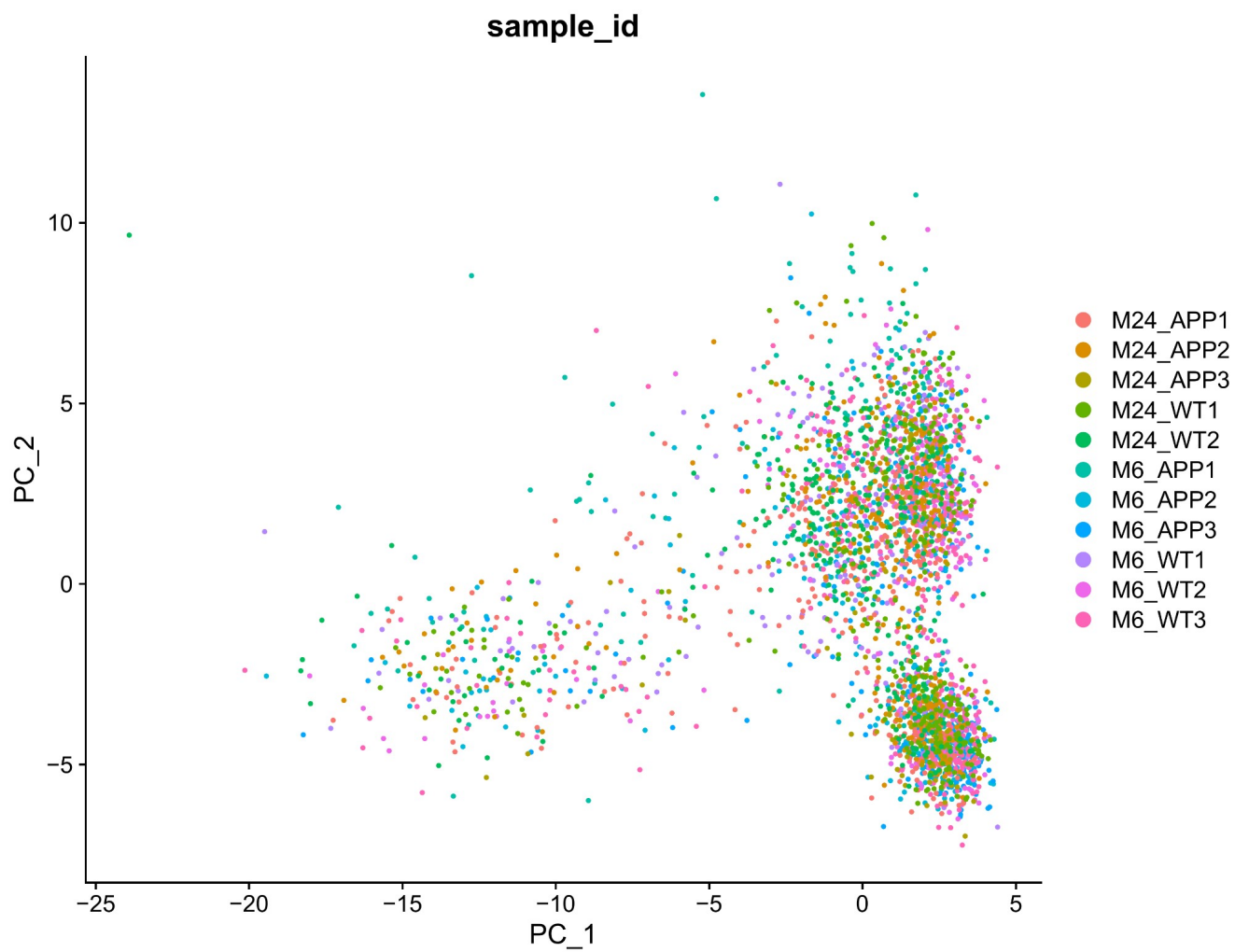
B. Selection of 2,000 highly variable genes.



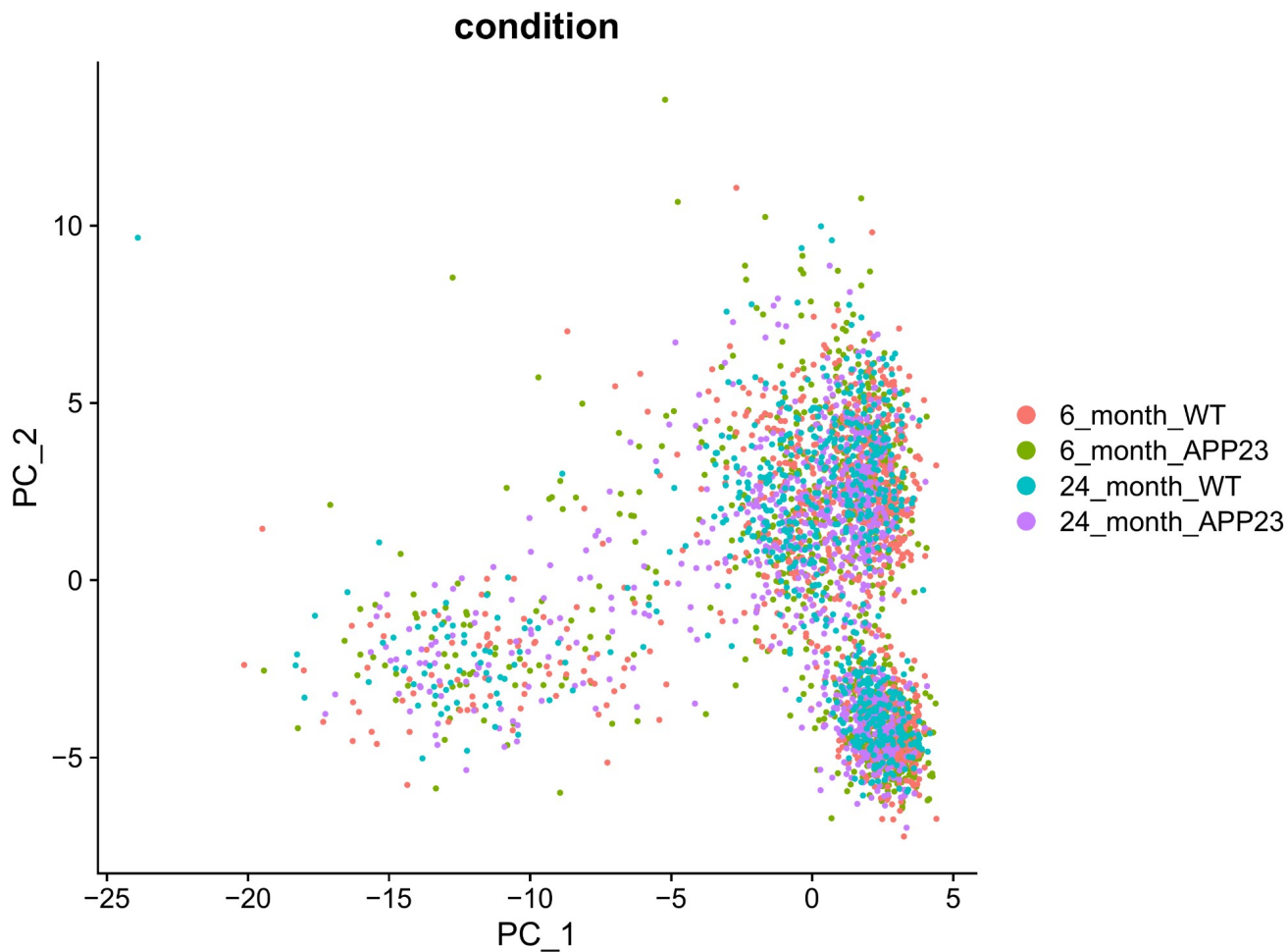
C. PCA elbow plot used to assess the number of principal components.



D. PCA colored by biological sample.



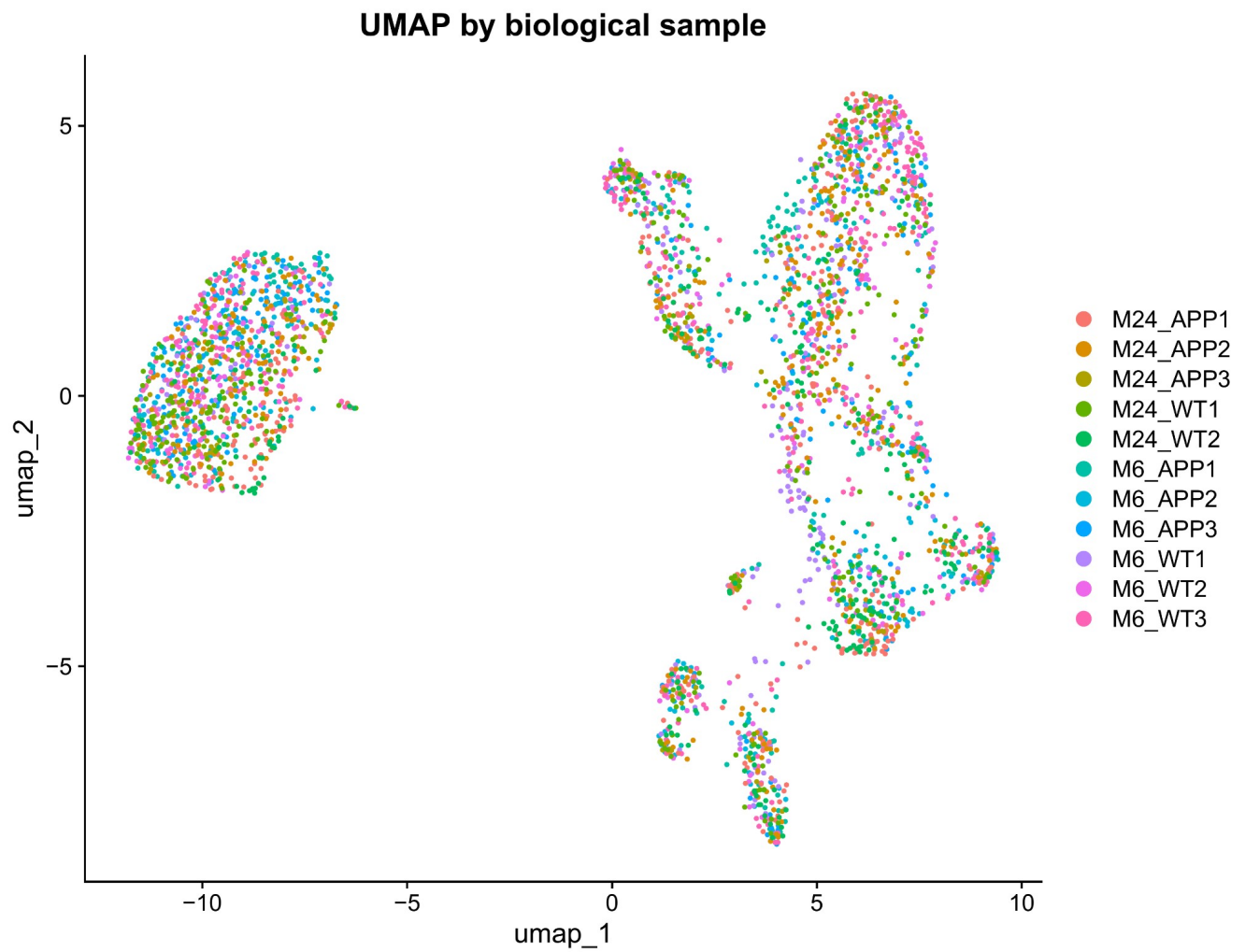
E. PCA colored by age/genotype condition.



Legend. (A) Per-sample distributions of detected genes, transcript counts, and ribosomal percentage. (B) Selection of 2,000 highly variable genes. (C) PCA elbow plot used to assess the number of principal components. (D) PCA colored by biological sample. (E) PCA colored by age/genotype condition.

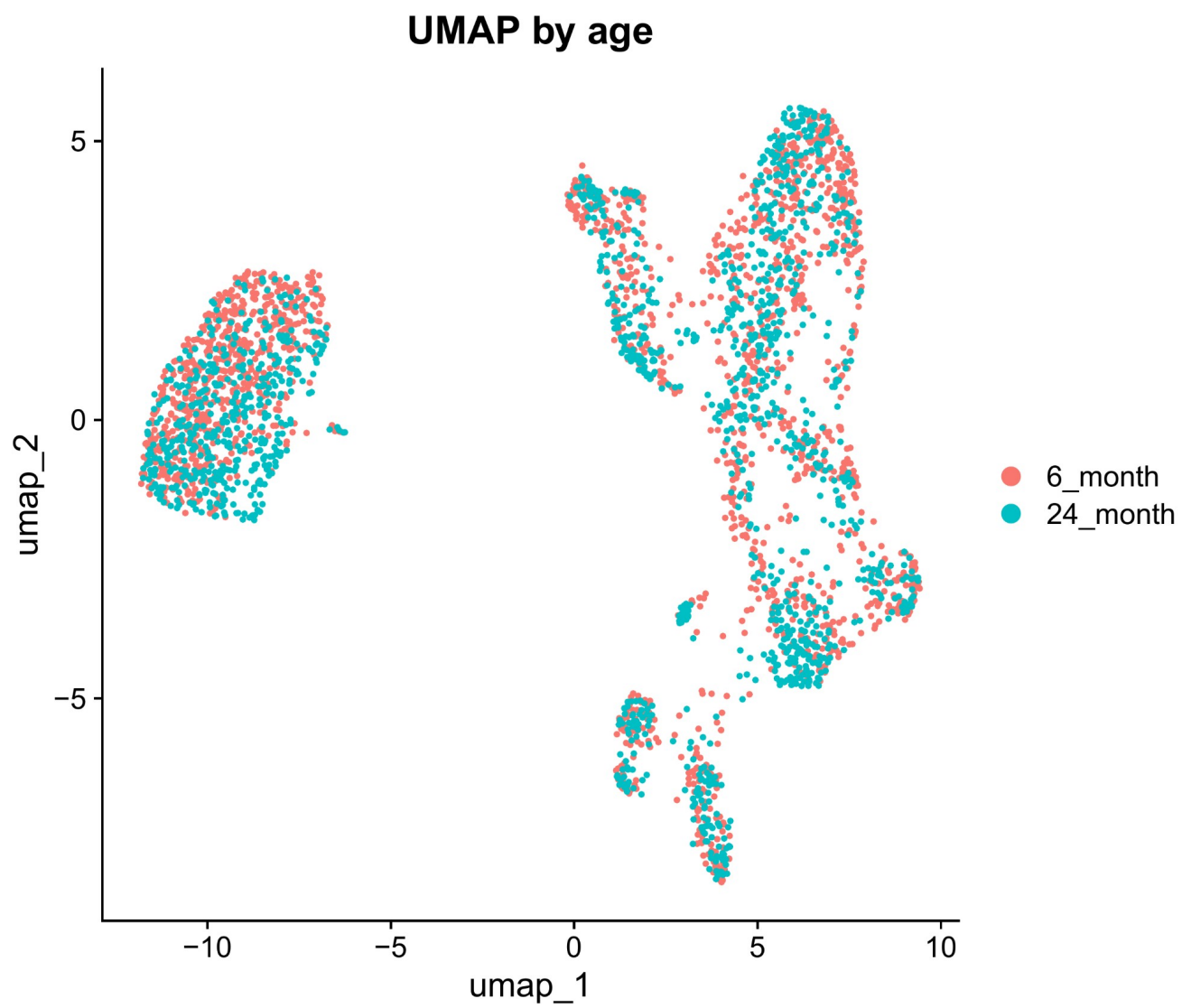
Supplementary Figure S2. Additional UMAP views and neuronal annotation diagnostics.

A. UMAP colored by sample.



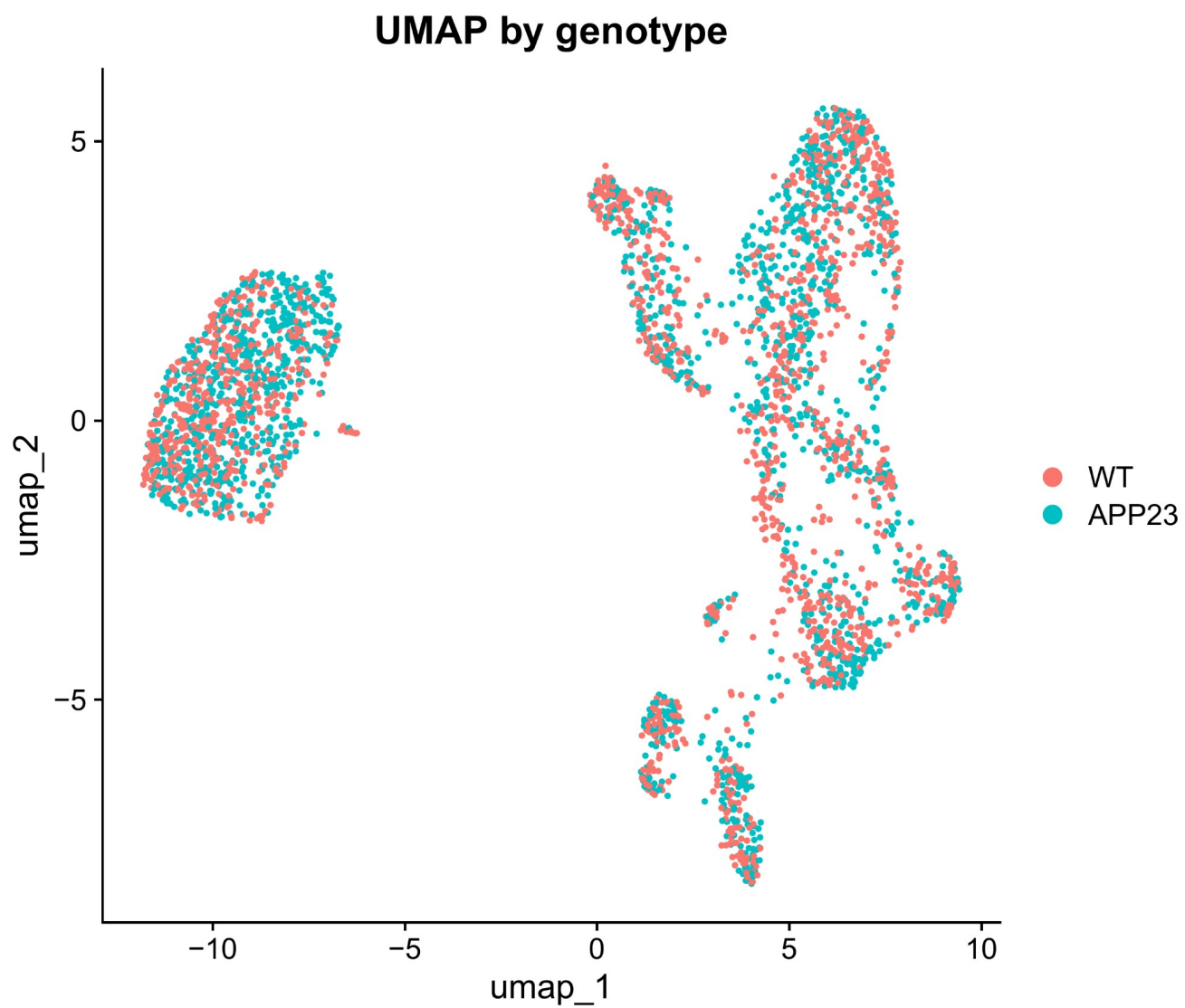
Supplementary Figure S2 (B)

B. UMAP colored by age.



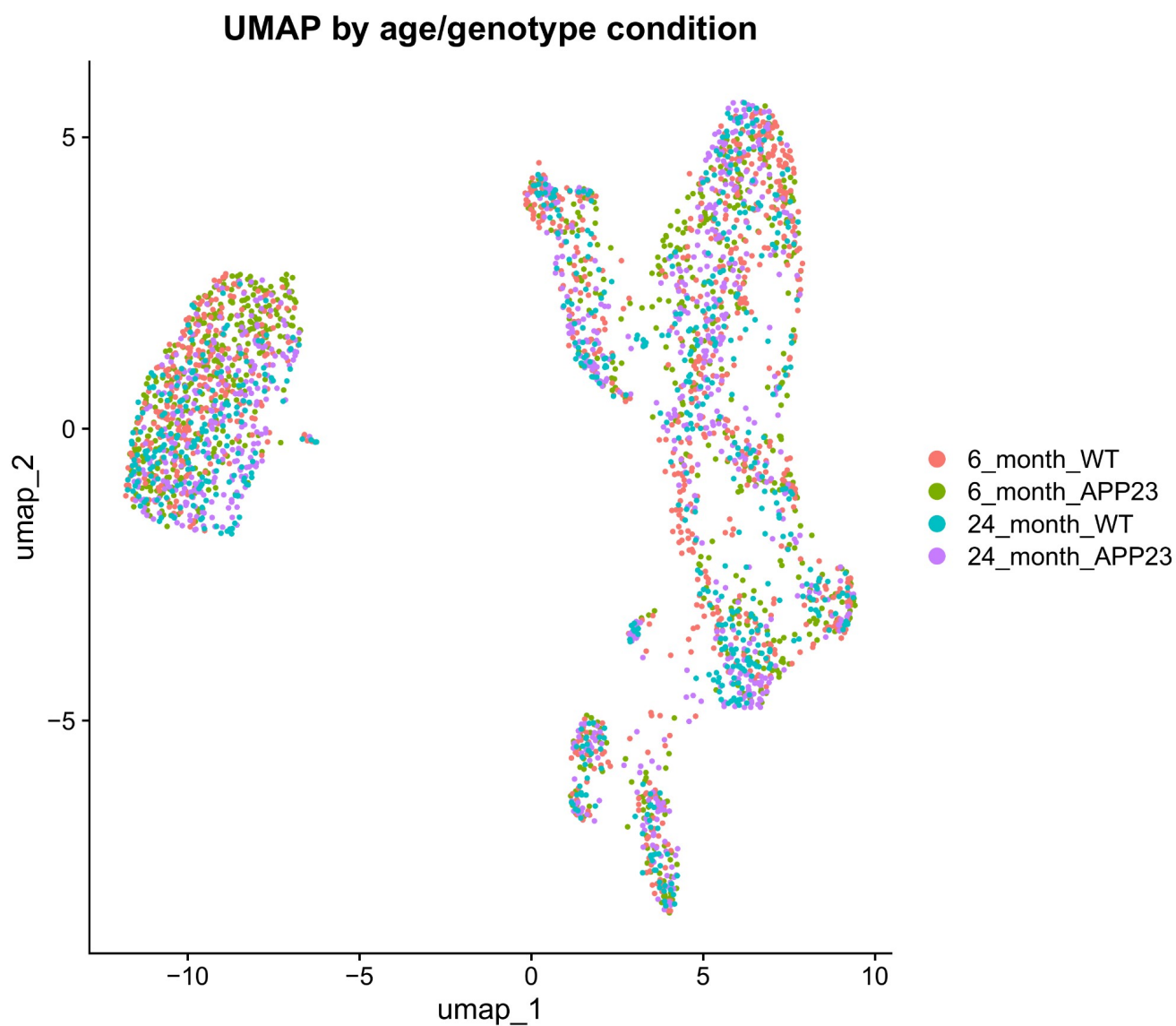
Supplementary Figure S2 (C)

C. UMAP colored by genotype.



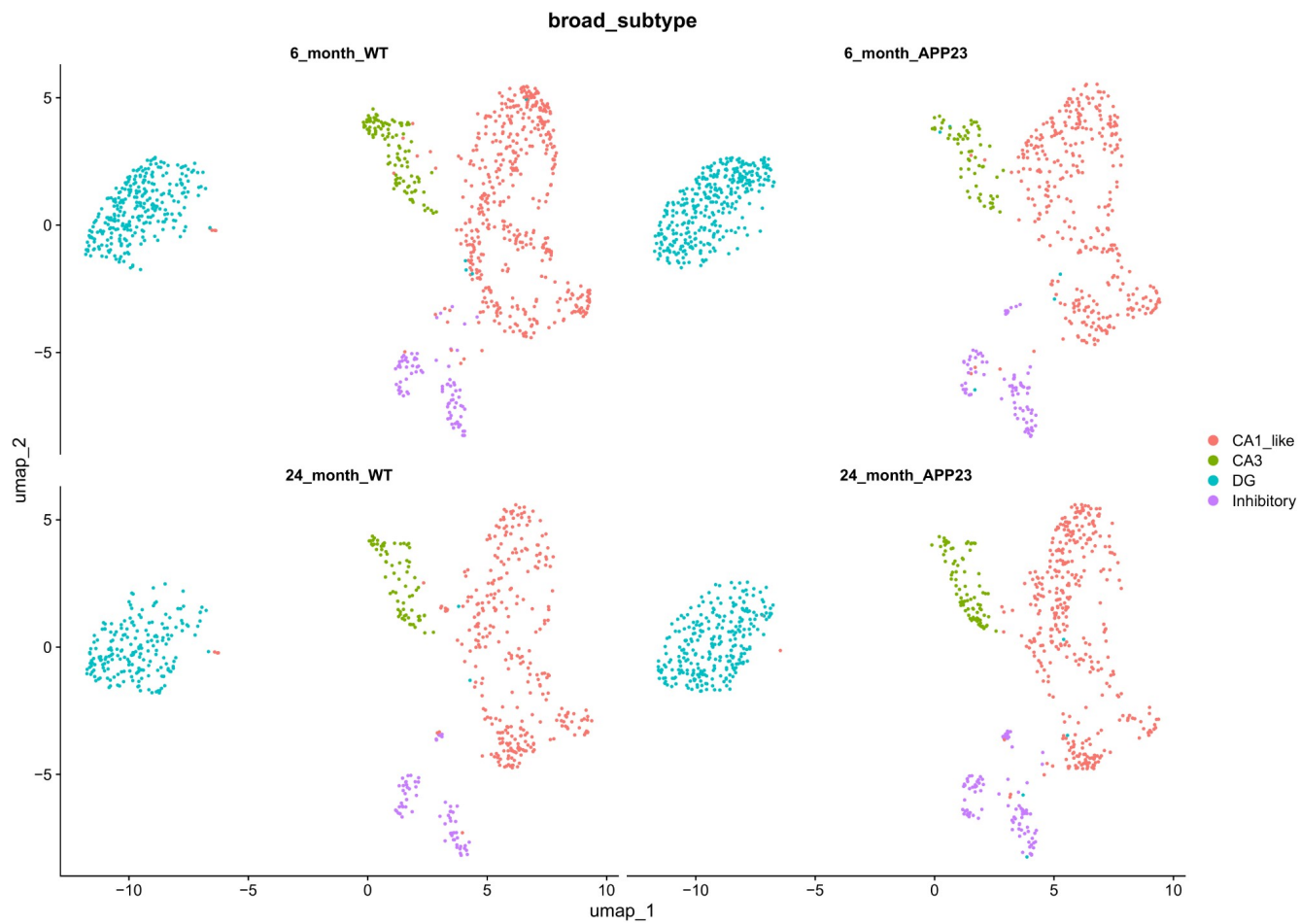
Supplementary Figure S2 (D)

D. UMAP colored by combined age/genotype condition.



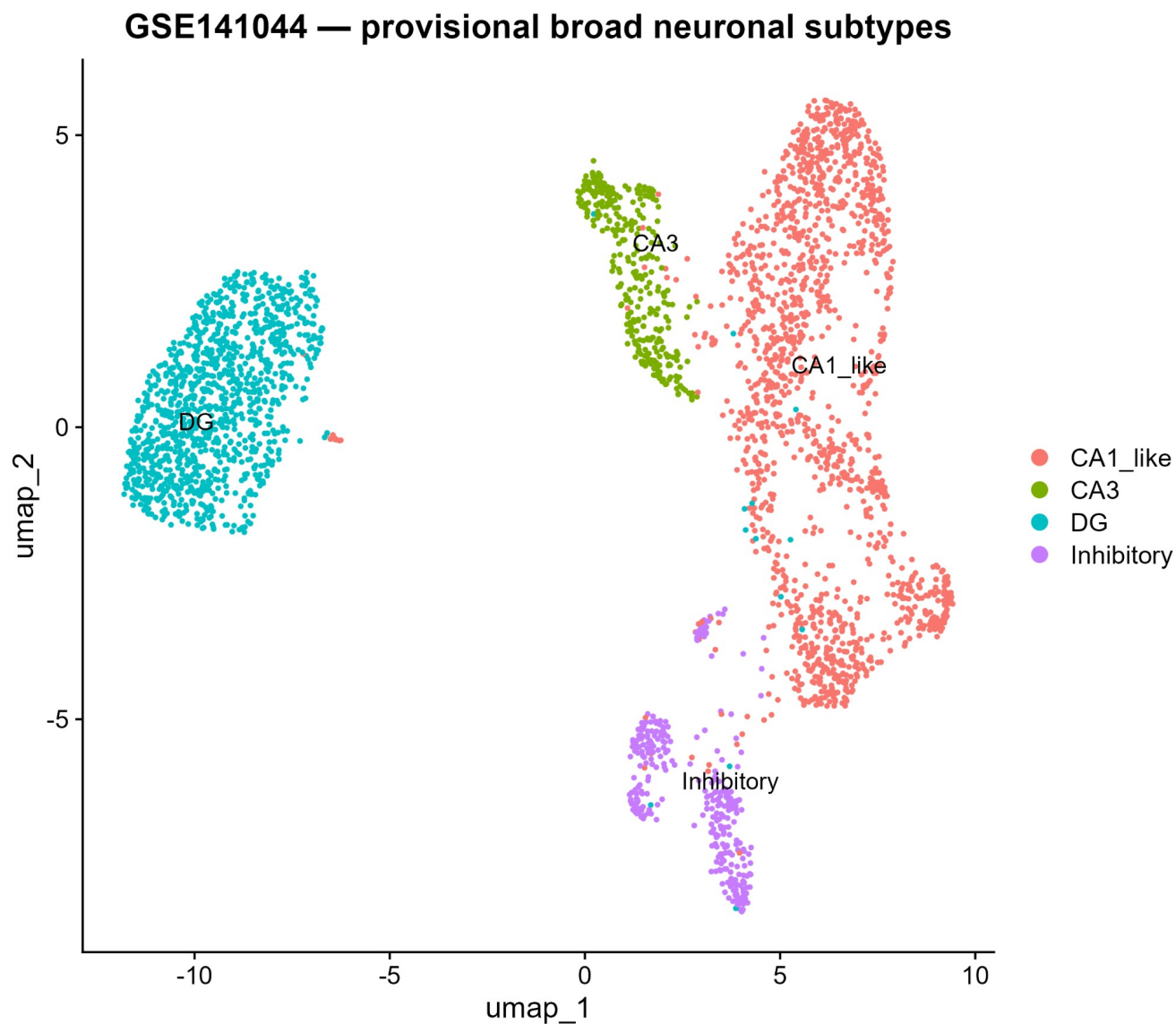
Supplementary Figure S2 (E)

E. Broad neuronal subtypes displayed by condition.



Supplementary Figure S2 (F)

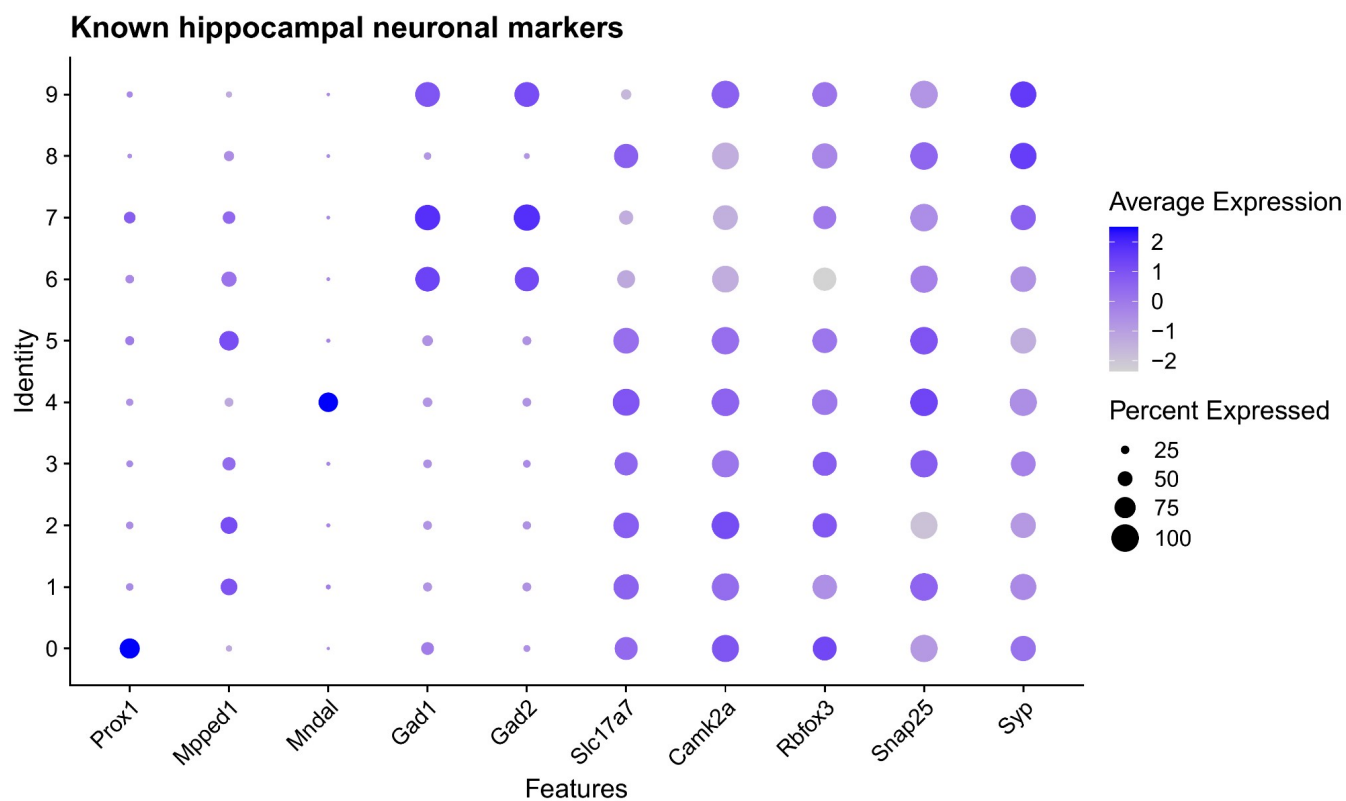
F. Provisional broad neuronal subtype annotation.



Legend. (A) UMAP colored by sample. (B) UMAP colored by age. (C) UMAP colored by genotype. (D) UMAP colored by combined age/genotype condition. (E) Broad neuronal subtypes displayed by condition. (F) Provisional broad neuronal subtype annotation.

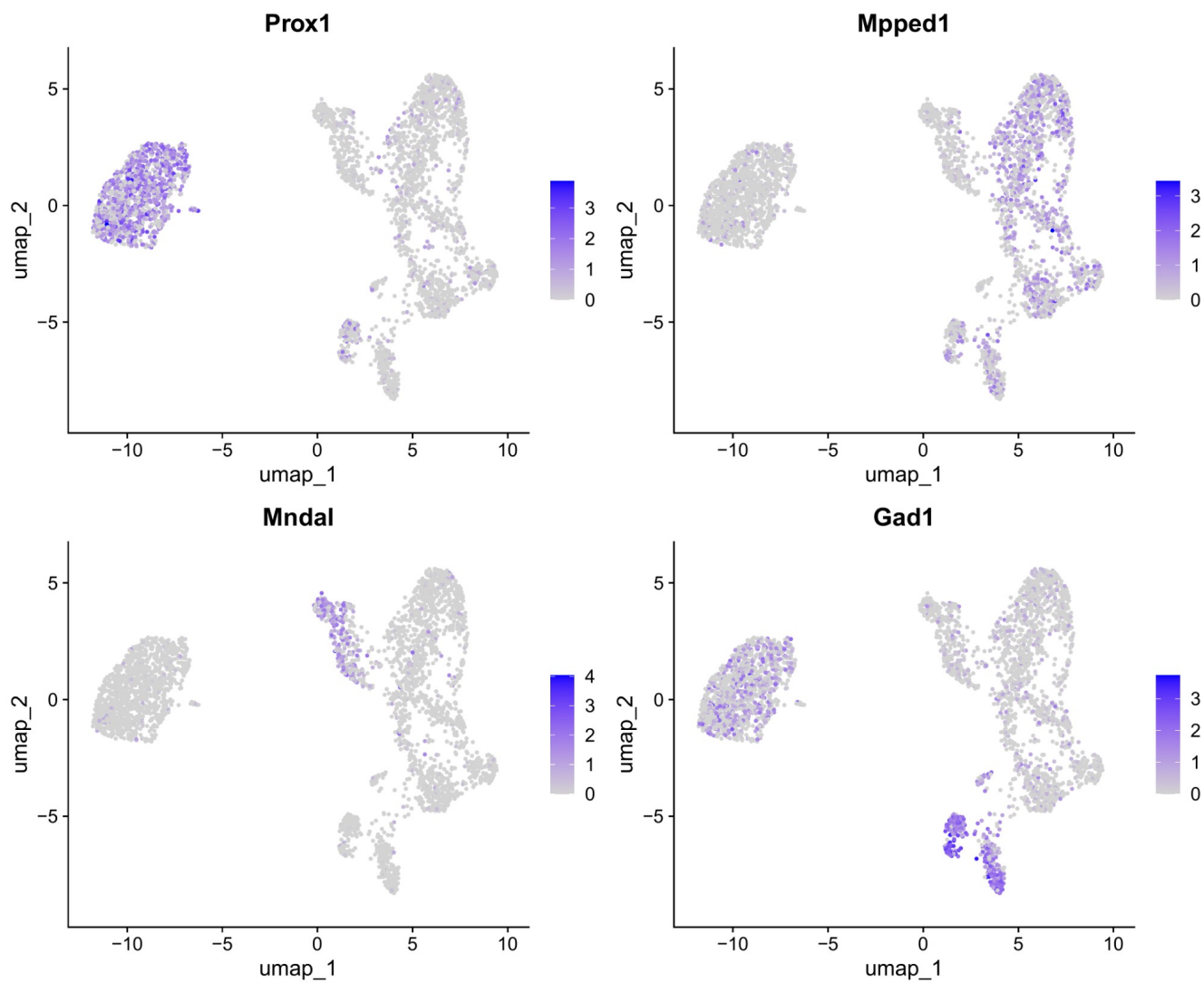
Supplementary Figure S3. Marker-based evaluation of neuronal cluster identities.

A. Dot plot of neuronal markers used in the published study.



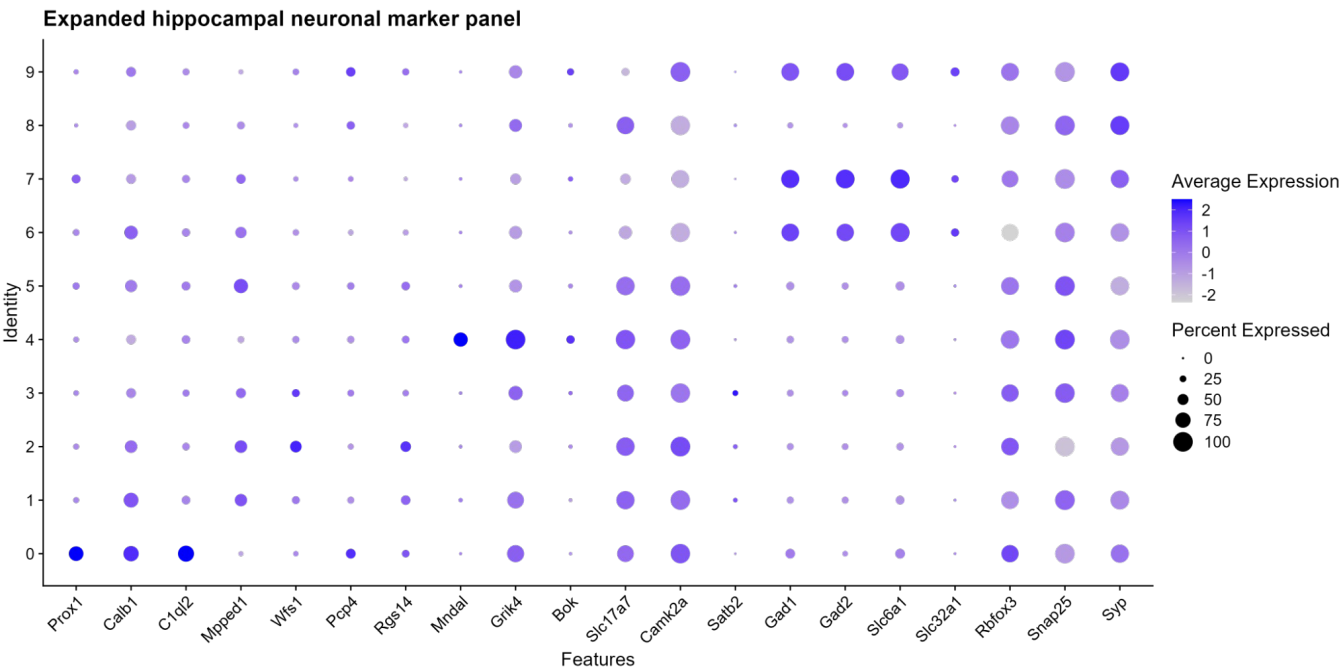
Supplementary Figure S3 (B)

B. Feature plots of published neuronal subtype markers.



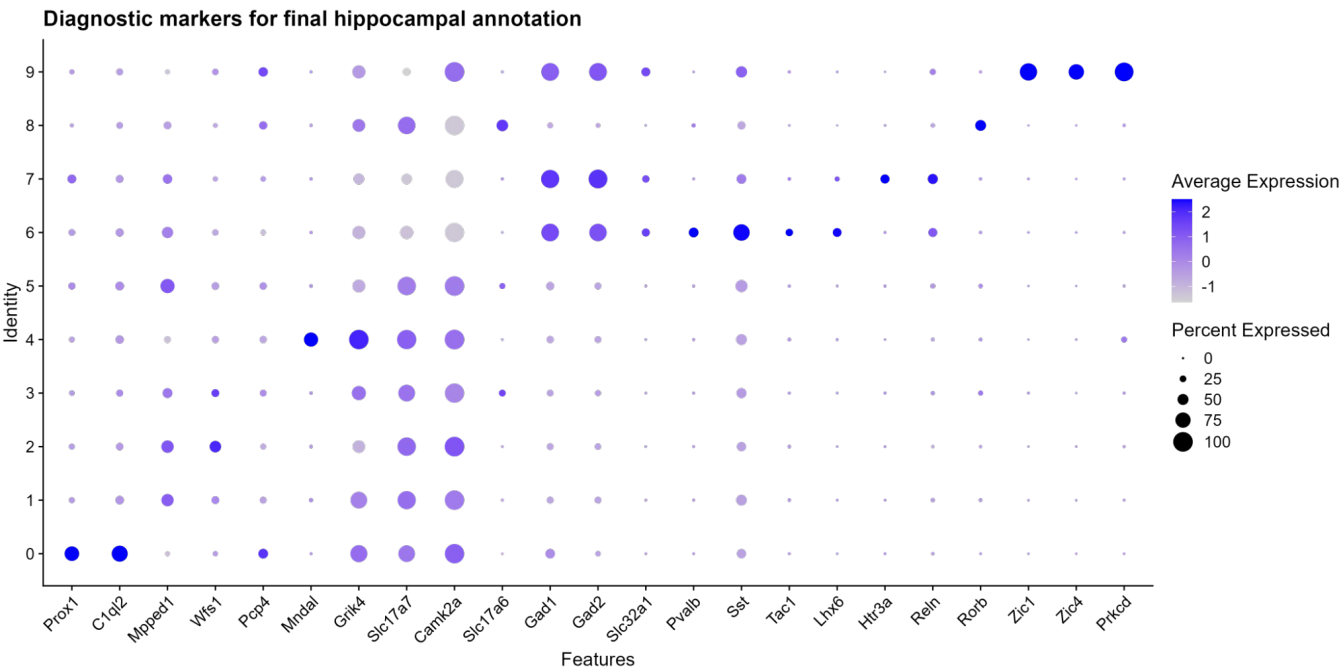
Supplementary Figure S3 (C)

C. Expanded hippocampal marker panel across clusters.



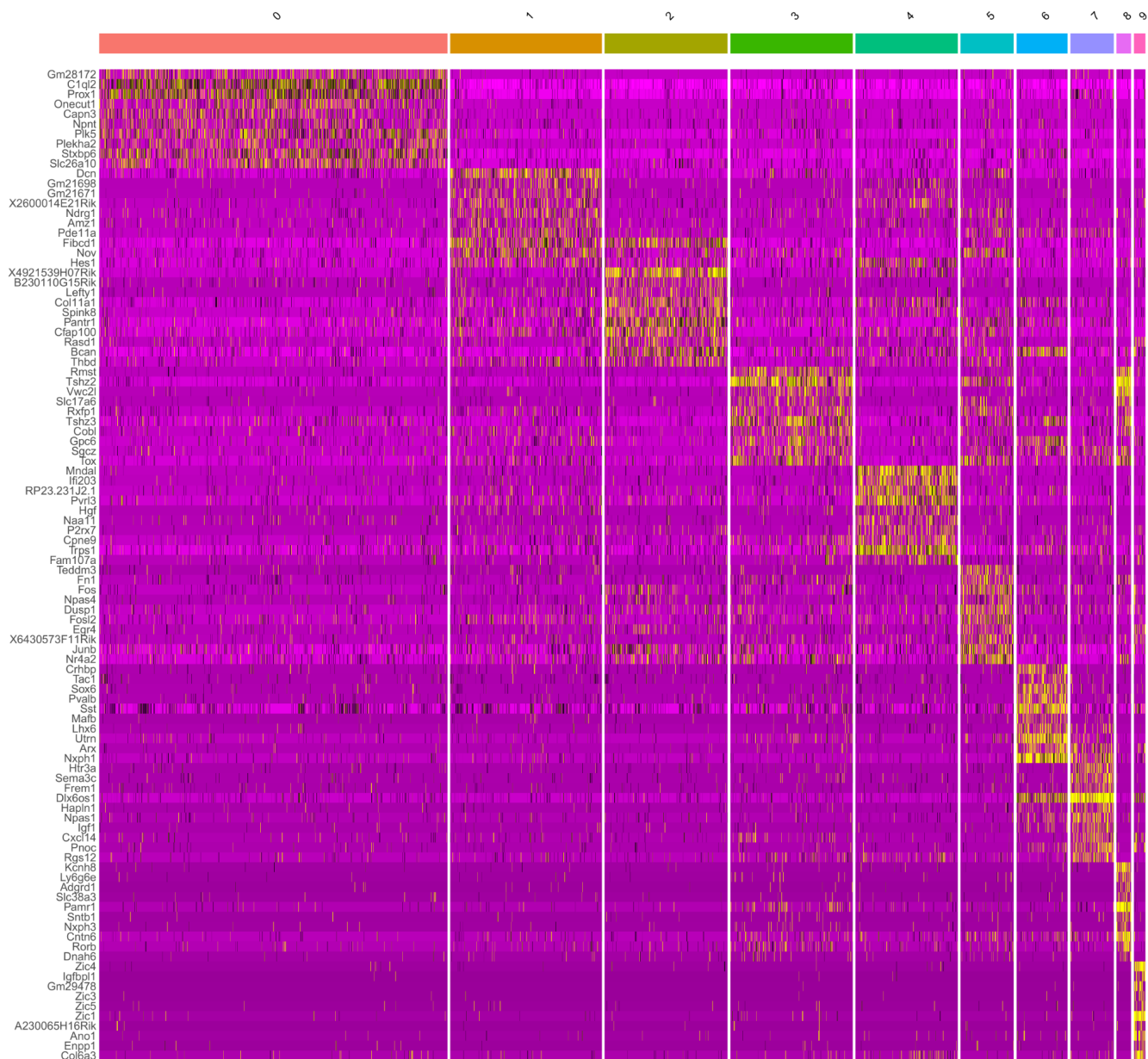
Supplementary Figure S3 (D)

D. Diagnostic marker expression supporting final broad annotations.



Supplementary Figure S3 (E)

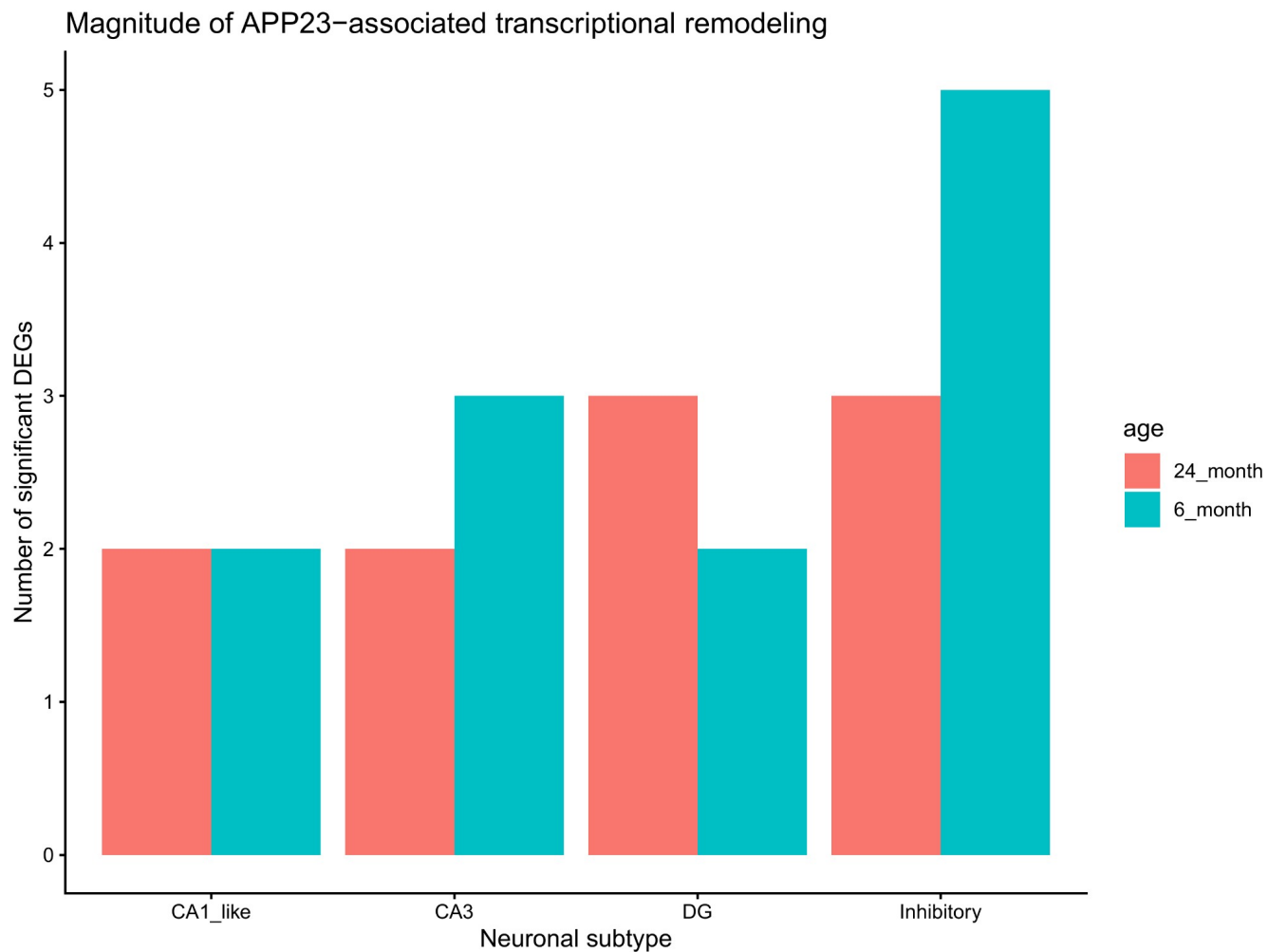
E. Heatmap of the top marker genes for the 10 fine clusters.



Legend. (A) Dot plot of neuronal markers used in the published study. (B) Feature plots of published neuronal subtype markers. (C) Expanded hippocampal marker panel across clusters. (D) Diagnostic marker expression supporting final broad annotations. (E) Heatmap of the top marker genes for the 10 fine clusters.

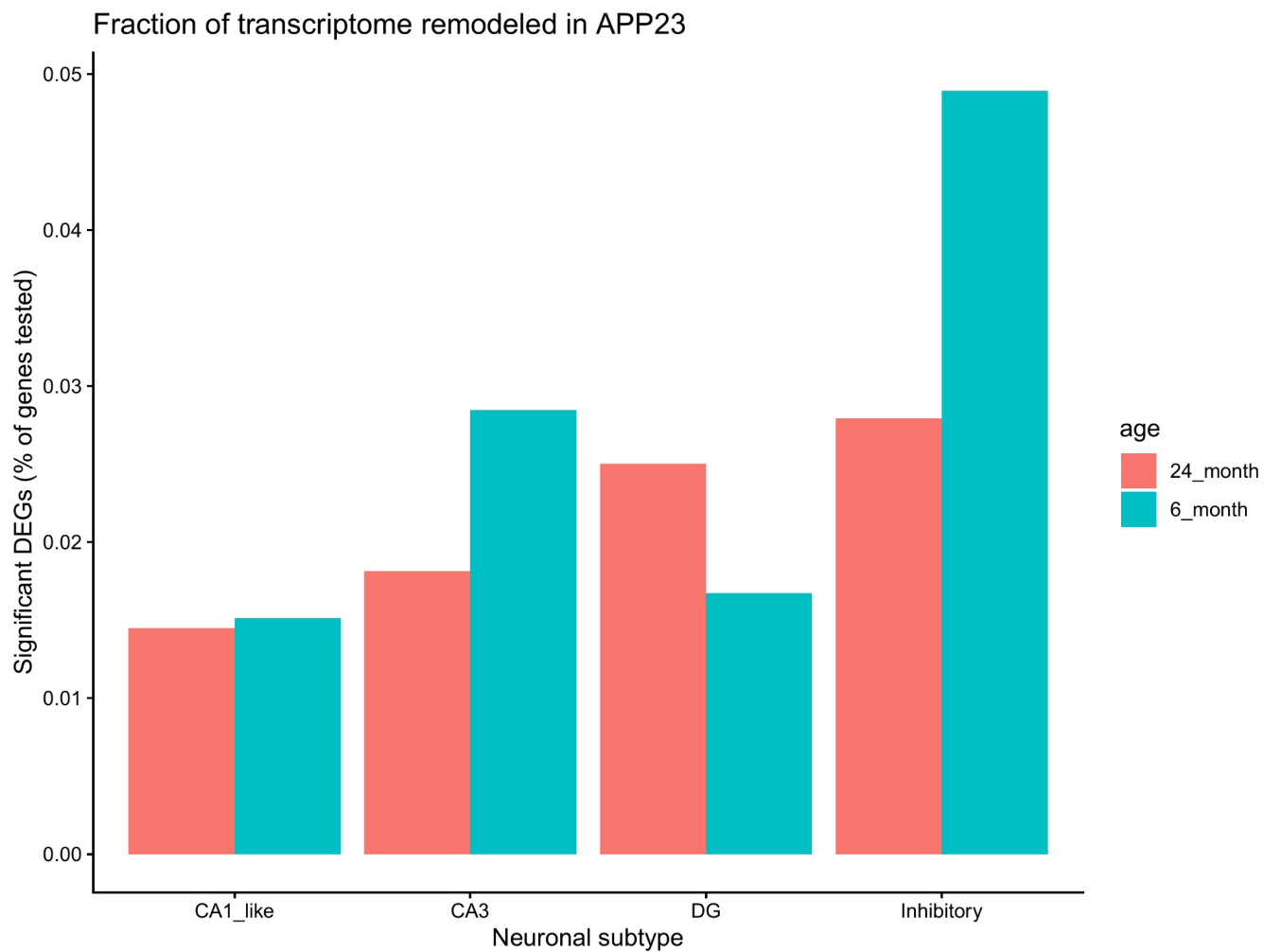
Supplementary Figure S4. Pseudobulk differential-expression summaries.

A. Number of significant differentially expressed genes across age and neuronal subtype comparisons.



Supplementary Figure S4 (B)

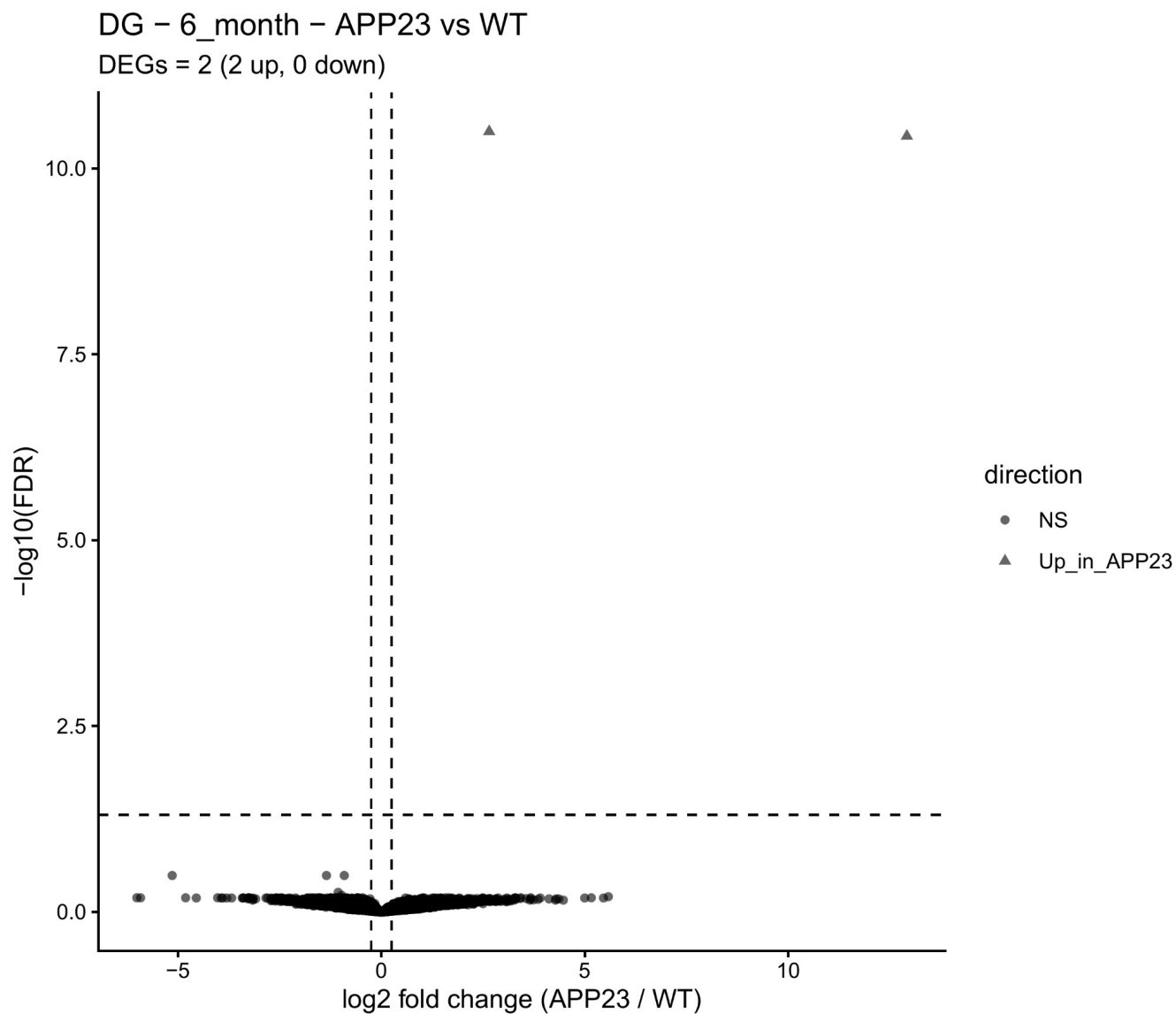
B. Percentage of tested genes meeting the differential-expression criteria.



Legend. (A) Number of significant differentially expressed genes across age and neuronal subtype comparisons. (B) Percentage of tested genes meeting the differential-expression criteria.

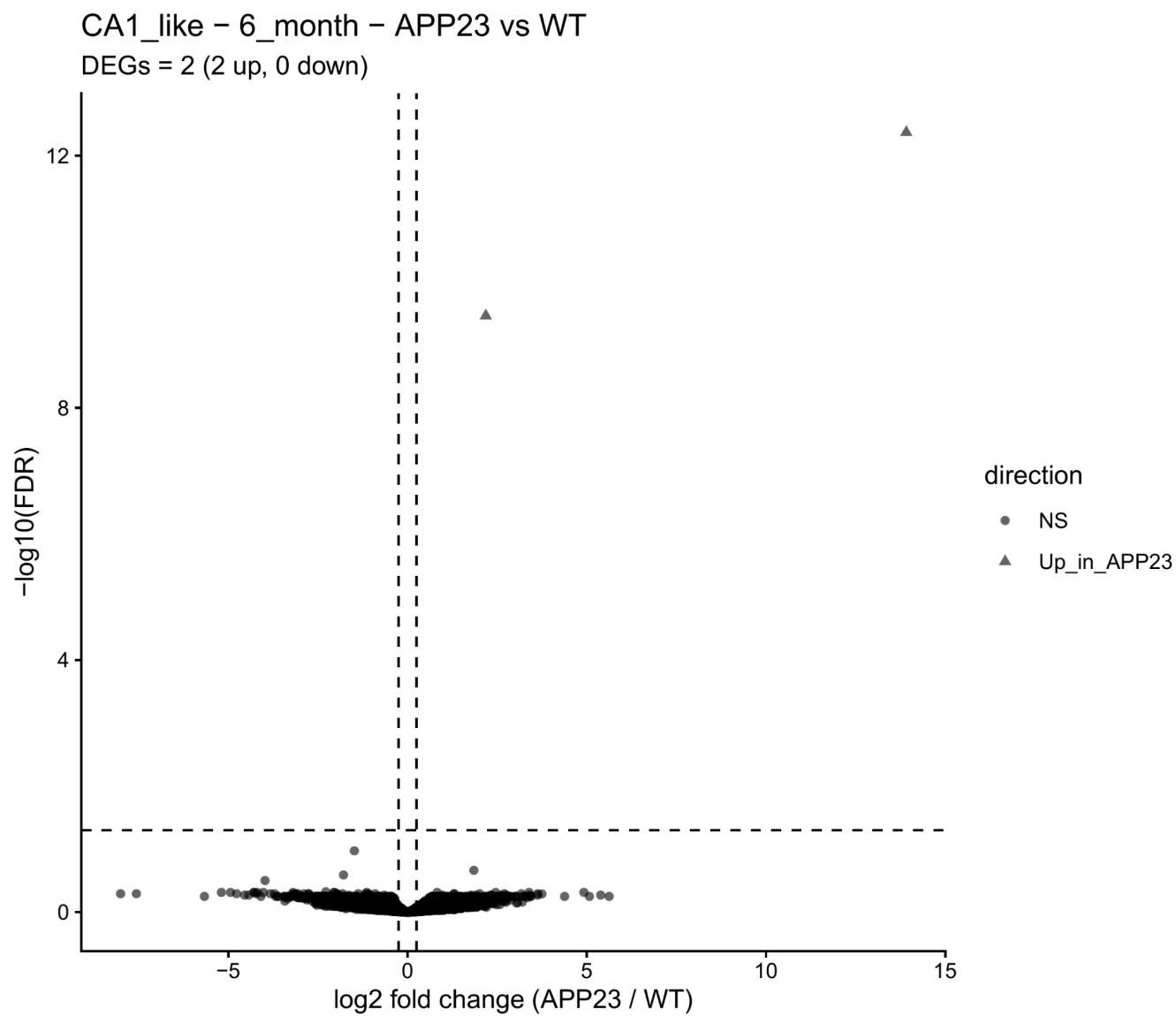
Supplementary Figure S5. Pseudobulk APP23-versus-WT differential expression at 6 months.

A. Dentate gyrus (DG).



Supplementary Figure S5 (B)

B. CA1-like neurons.

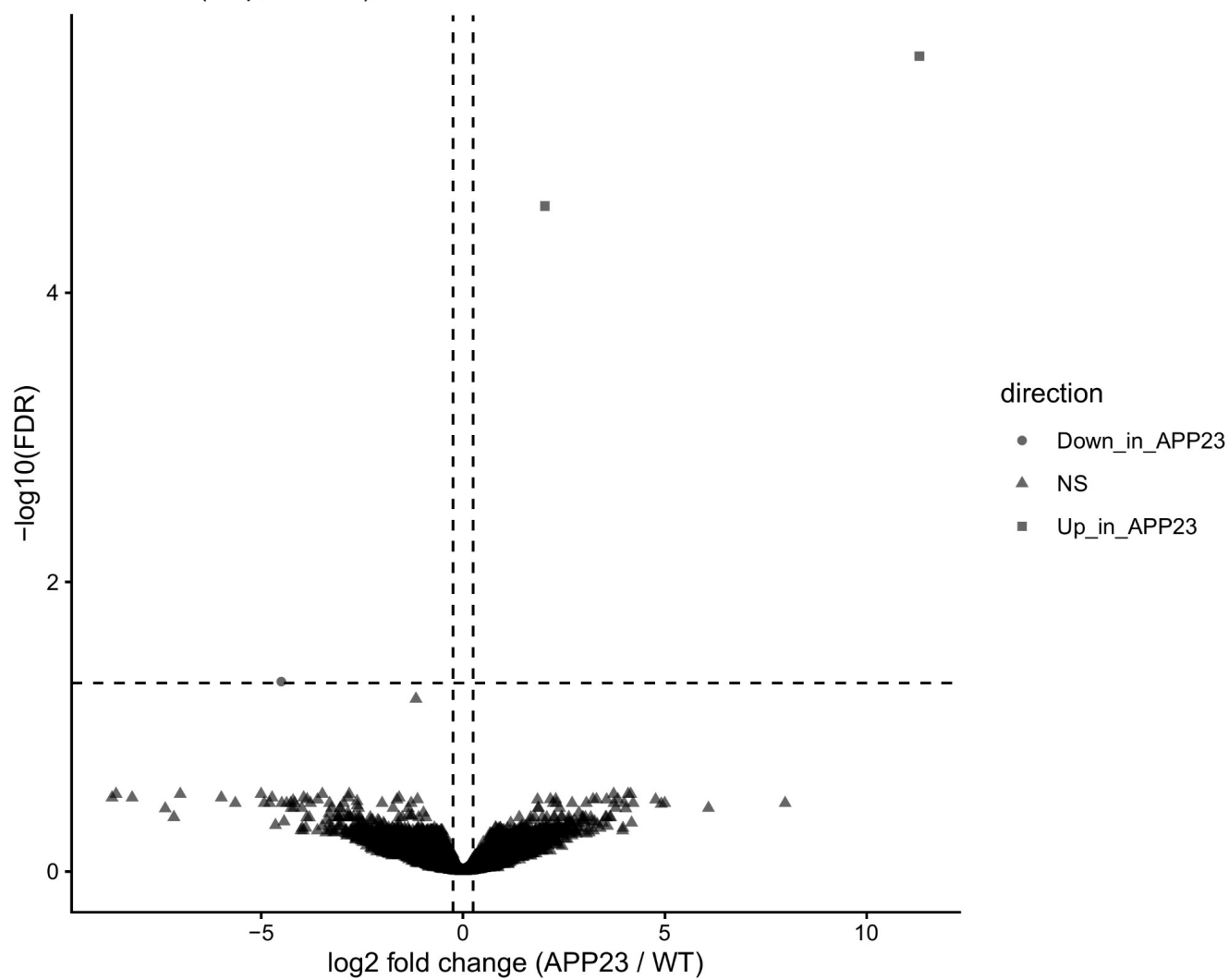


Supplementary Figure S5 (C)

C. CA3 neurons.

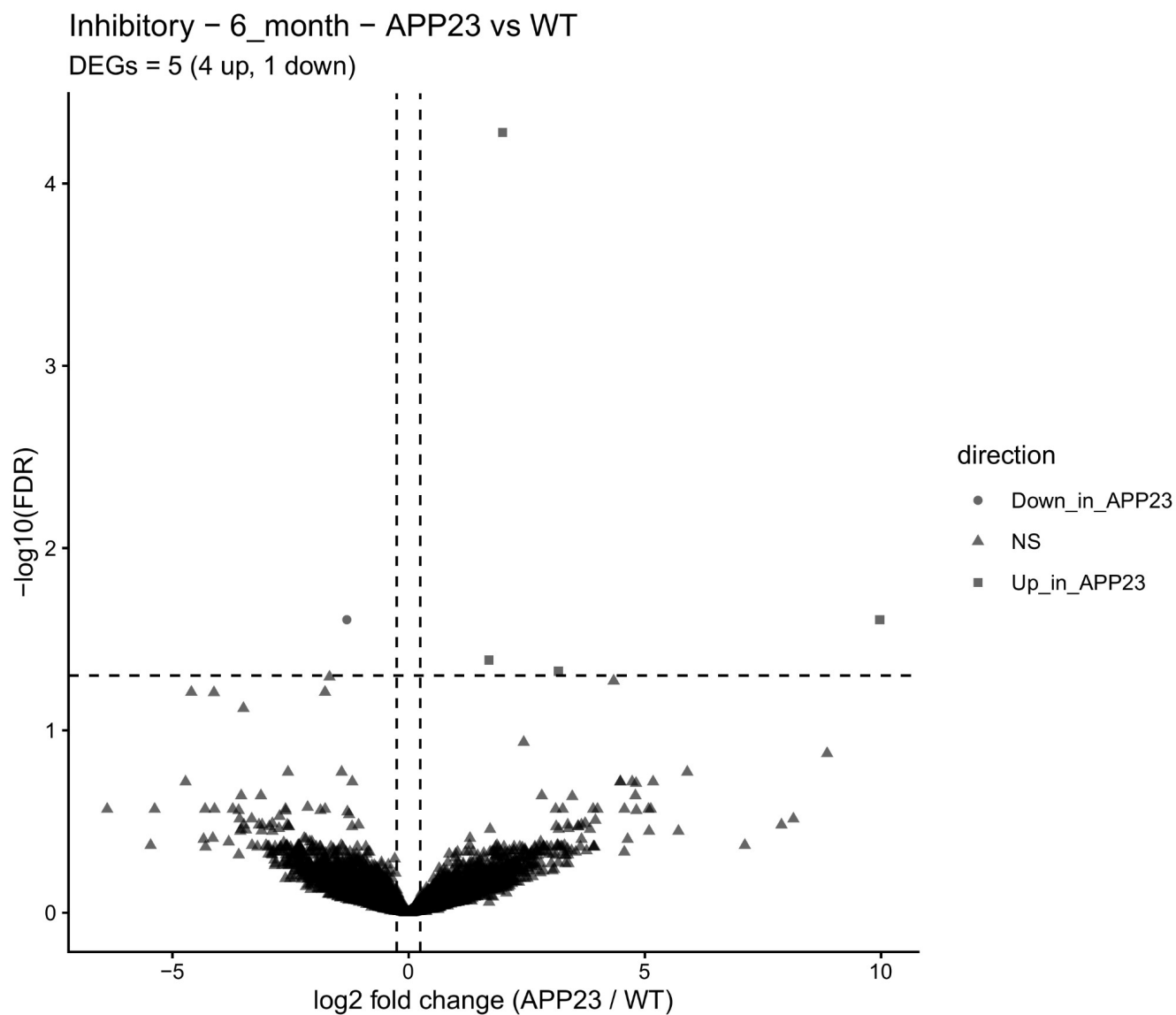
CA3 – 6_month – APP23 vs WT

DEGs = 3 (2 up, 1 down)



Supplementary Figure S5 (D)

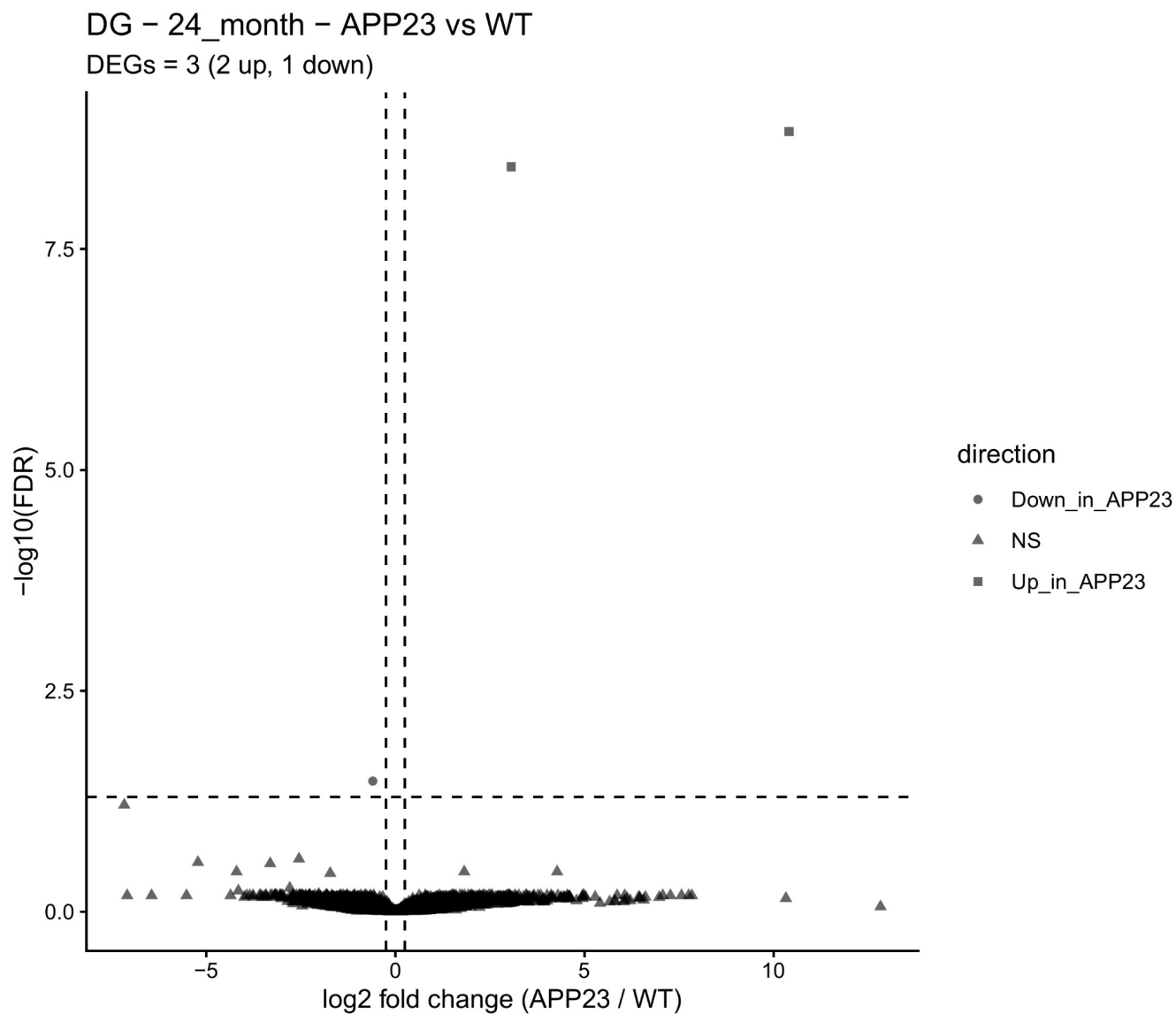
D. Inhibitory neurons.



Legend. (A) Dentate gyrus (DG). (B) CA1-like neurons. (C) CA3 neurons. (D) Inhibitory neurons.

Supplementary Figure S6. Pseudobulk APP23-versus-WT differential expression at 24 months.

A. Dentate gyrus (DG).

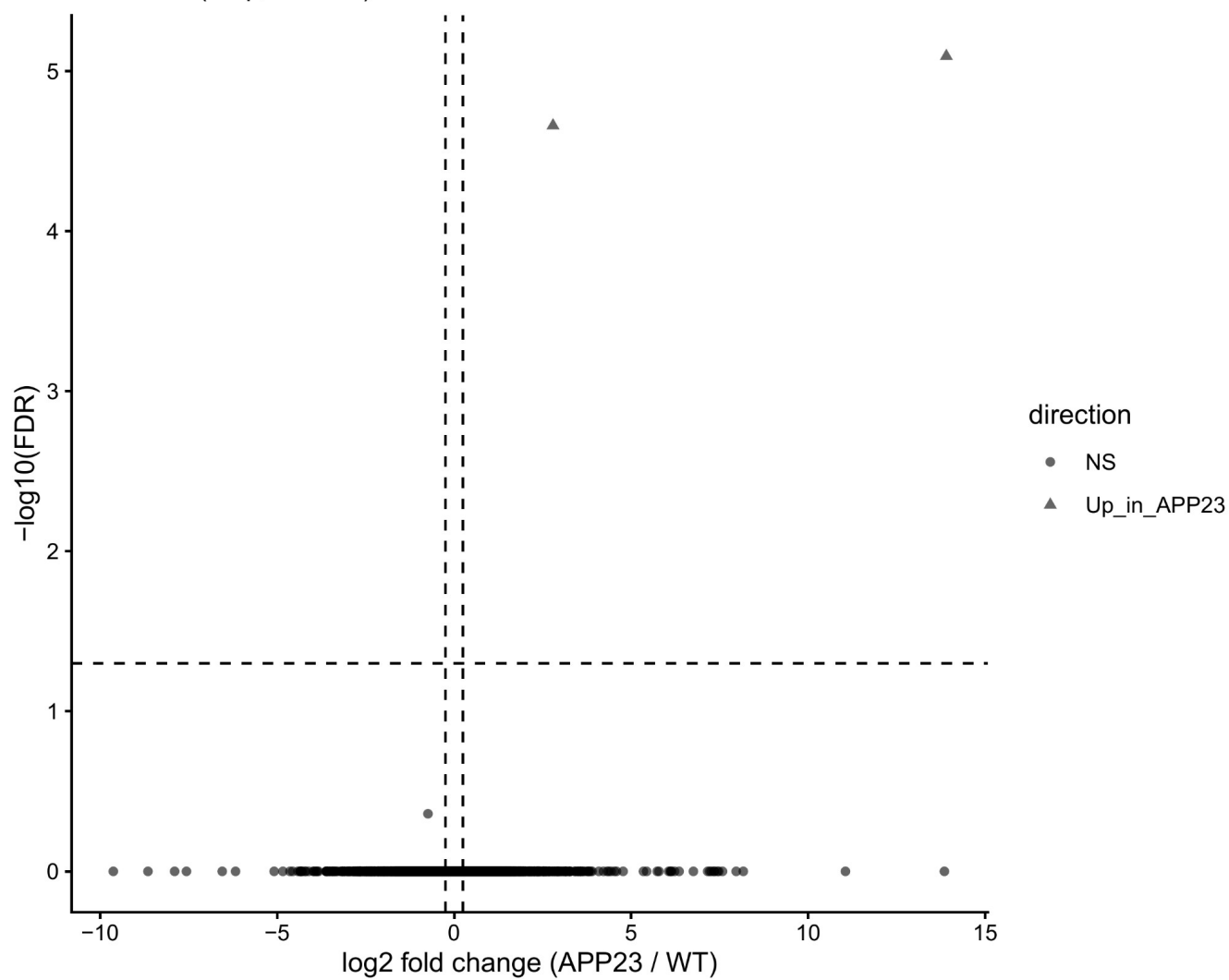


Supplementary Figure S6 (B)

B. CA1-like neurons.

CA1_like - 24_month - APP23 vs WT

DEGs = 2 (2 up, 0 down)

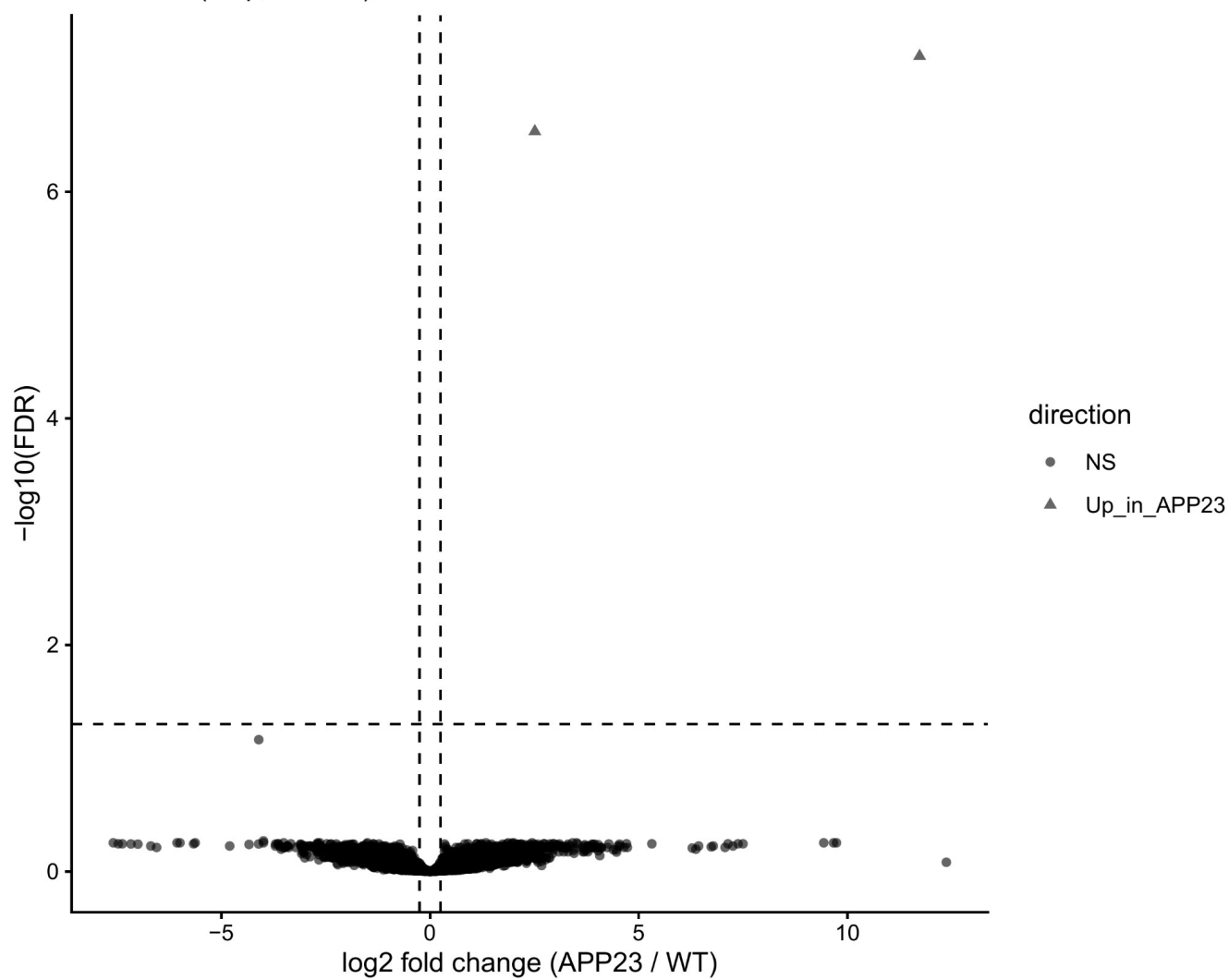


Supplementary Figure S6 (C)

C. CA3 neurons.

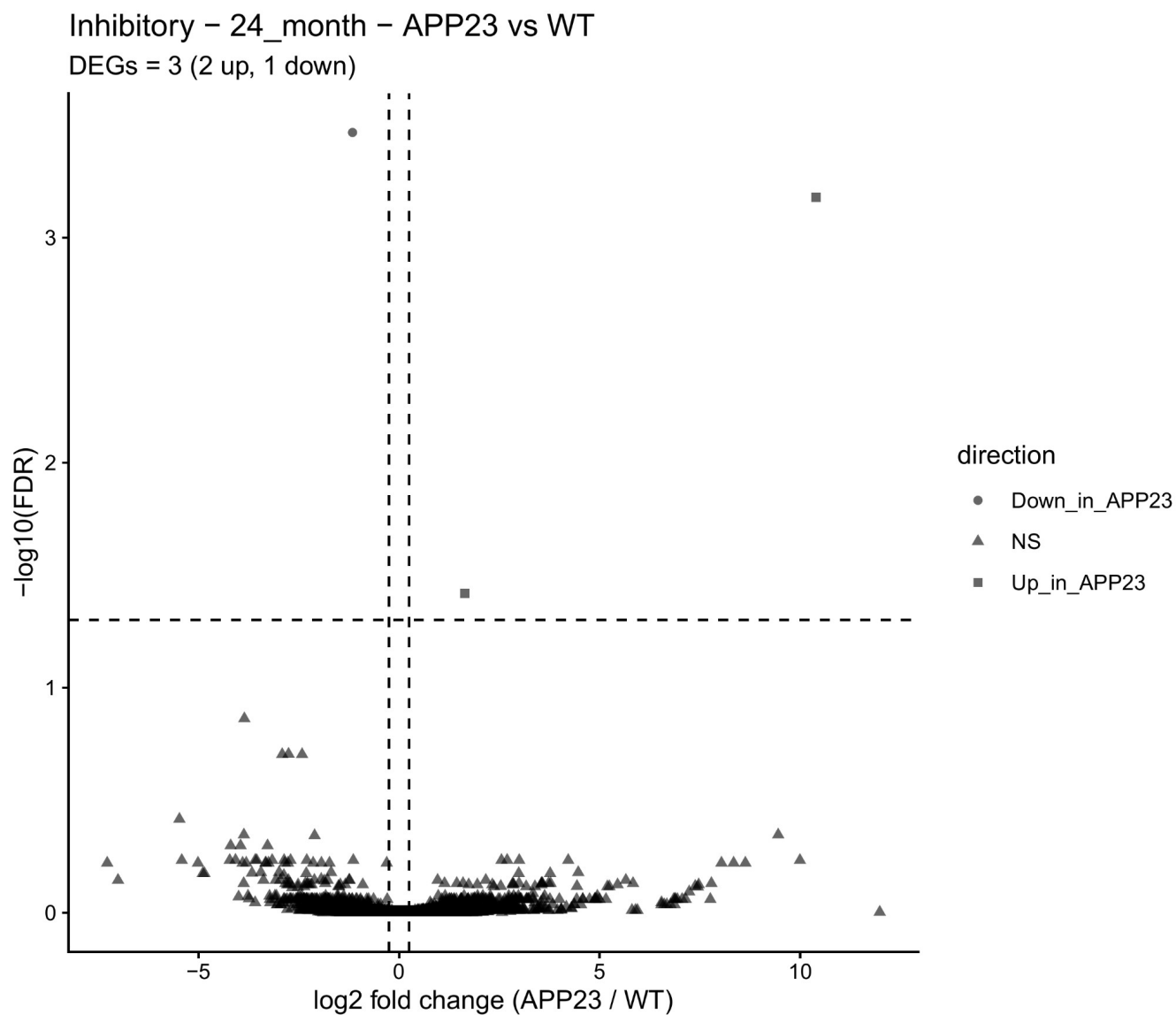
CA3 – 24_month – APP23 vs WT

DEGs = 2 (2 up, 0 down)



Supplementary Figure S6 (D)

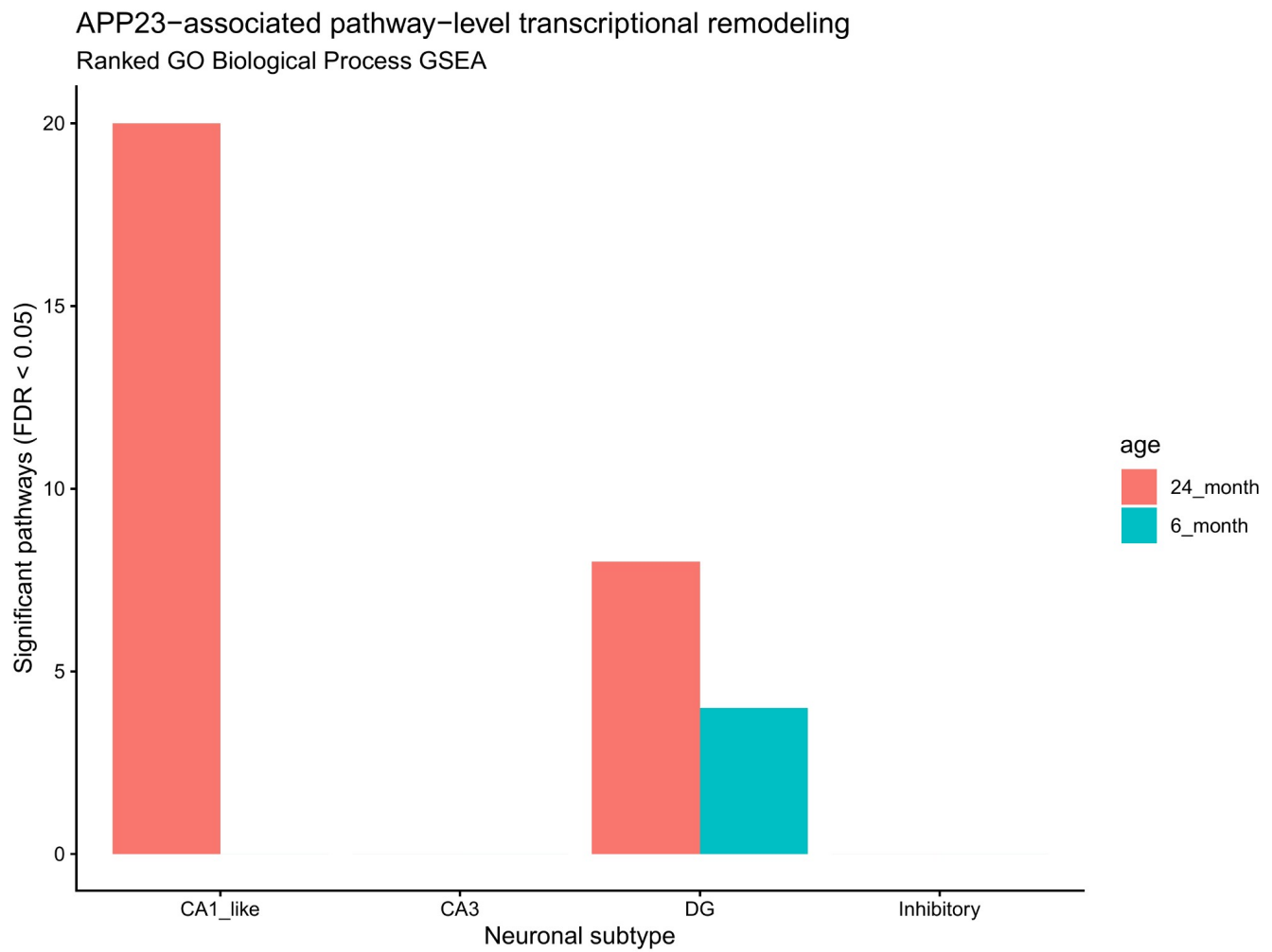
D. Inhibitory neurons.



Legend. (A) Dentate gyrus (DG). (B) CA1-like neurons. (C) CA3 neurons. (D) Inhibitory neurons.

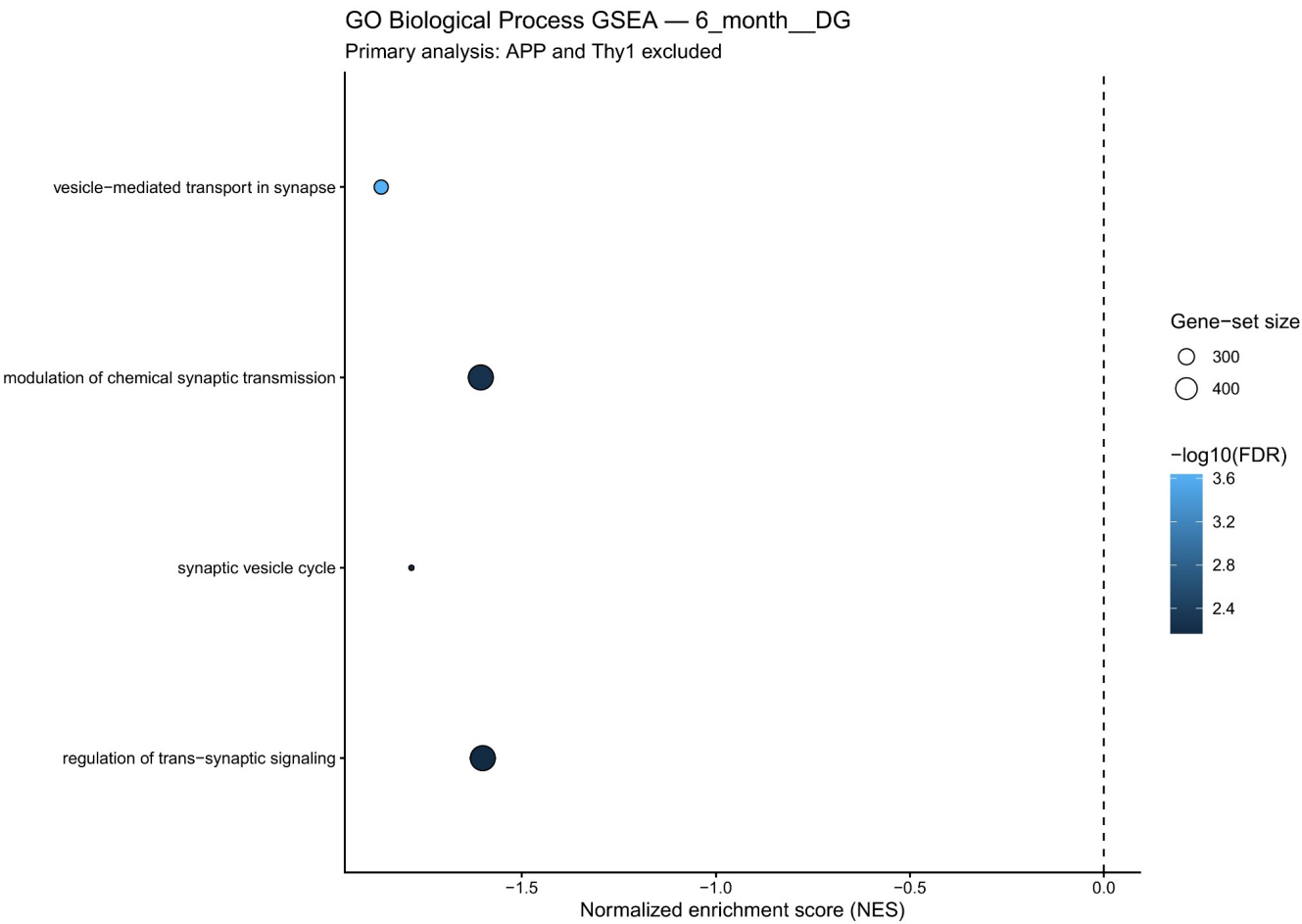
Supplementary Figure S7. Gene-set enrichment analysis of APP23-associated pathway remodeling.

A. Number of significant GO Biological Process pathways across comparisons.



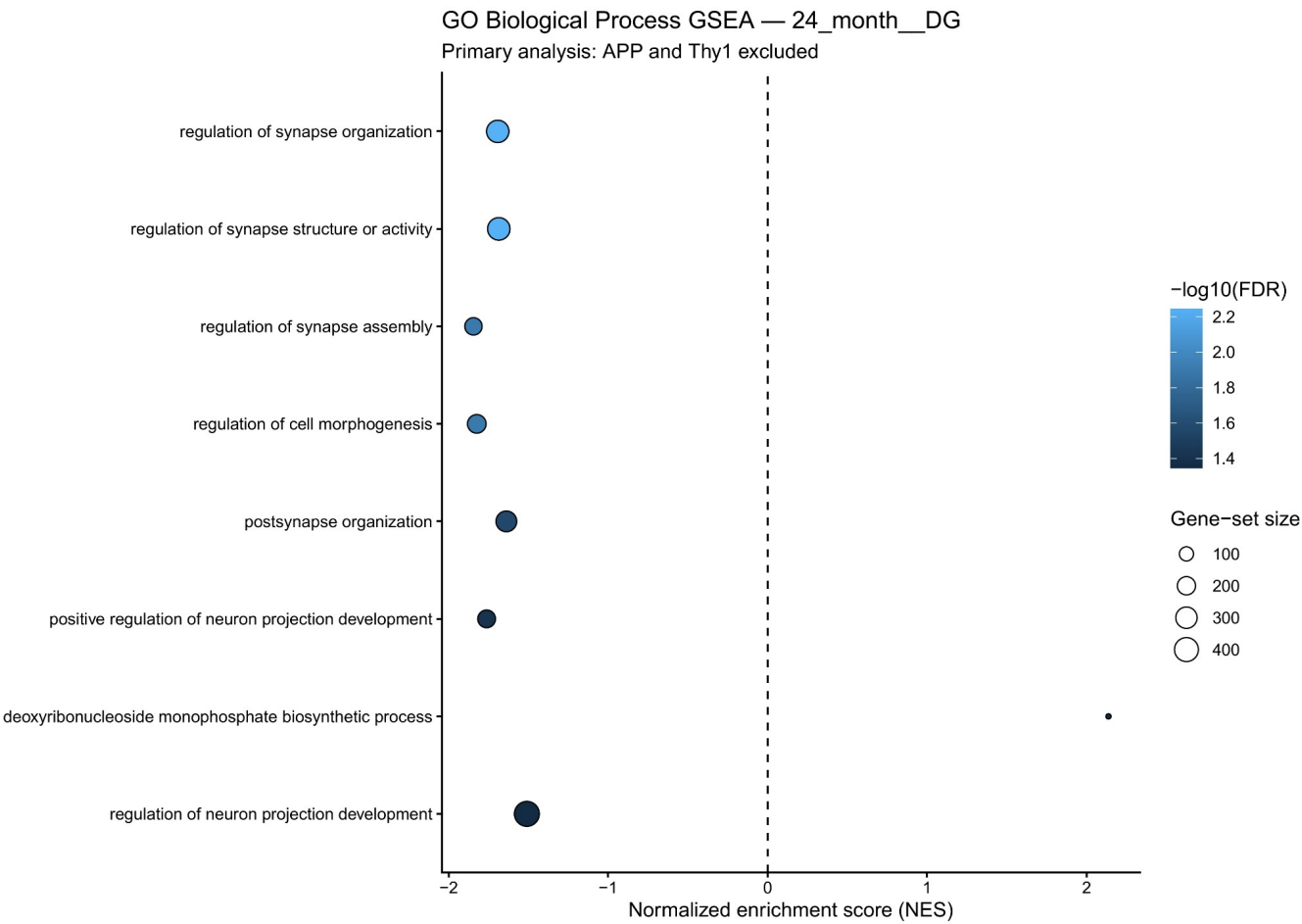
Supplementary Figure S7 (B)

B. Significant GO Biological Process enrichment in 6-month DG.



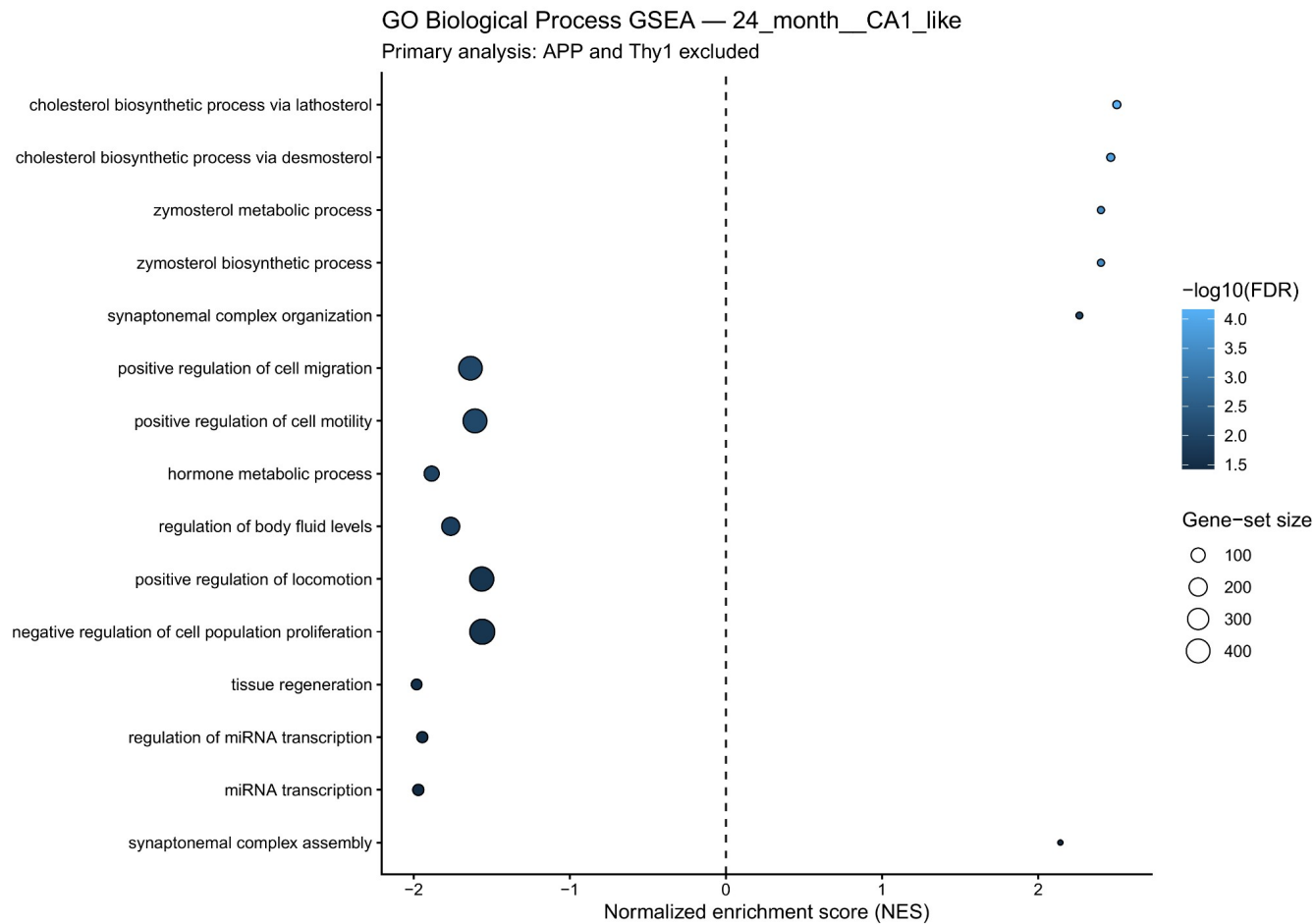
Supplementary Figure S7 (C)

C. Significant GO Biological Process enrichment in 24-month DG.



Supplementary Figure S7 (D)

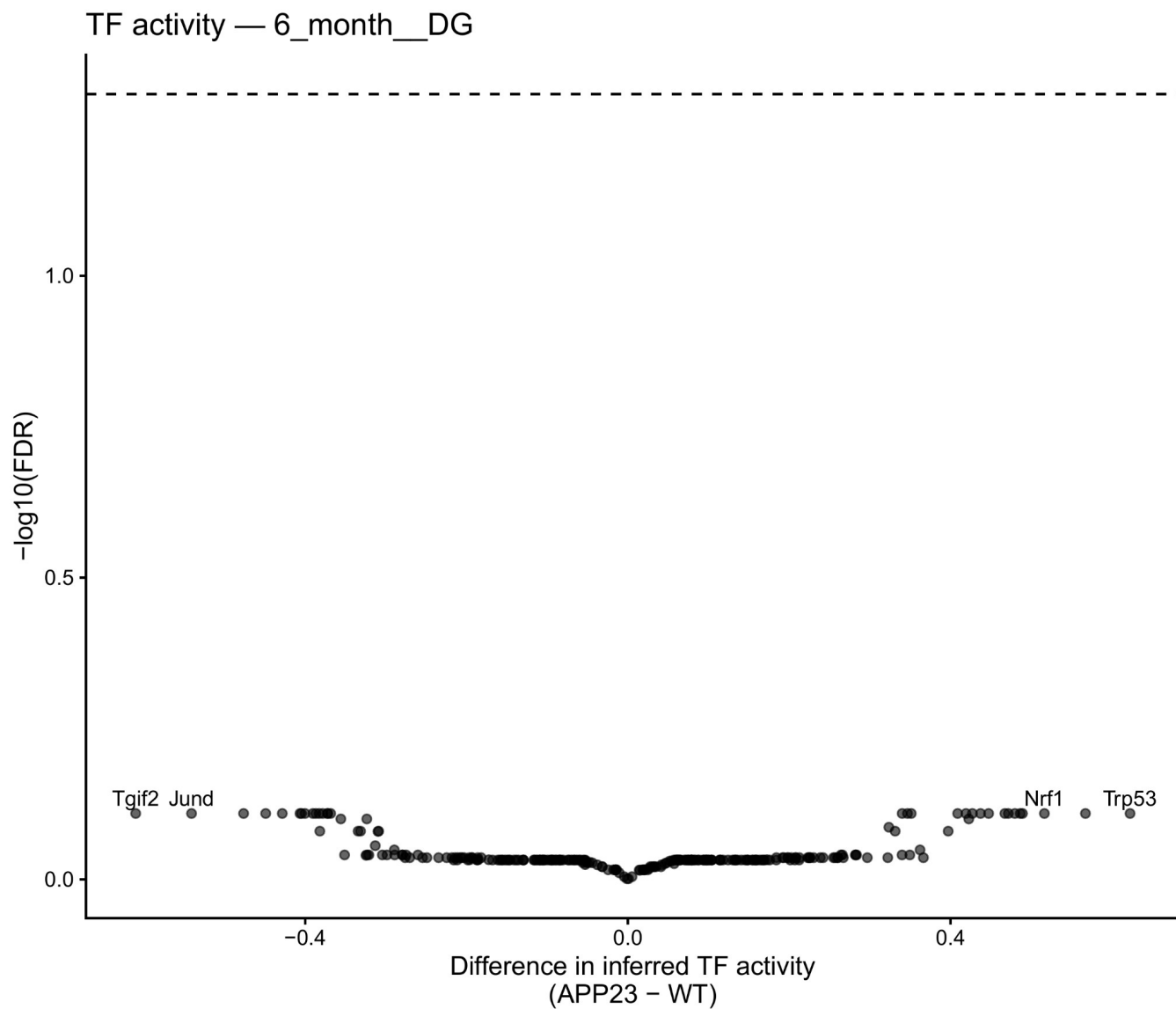
D. Significant GO Biological Process enrichment in 24-month CA1-like neurons.



Legend. (A) Number of significant GO Biological Process pathways across comparisons. (B) Significant GO Biological Process enrichment in 6-month DG. (C) Significant GO Biological Process enrichment in 24-month DG. (D) Significant GO Biological Process enrichment in 24-month CA1-like neurons.

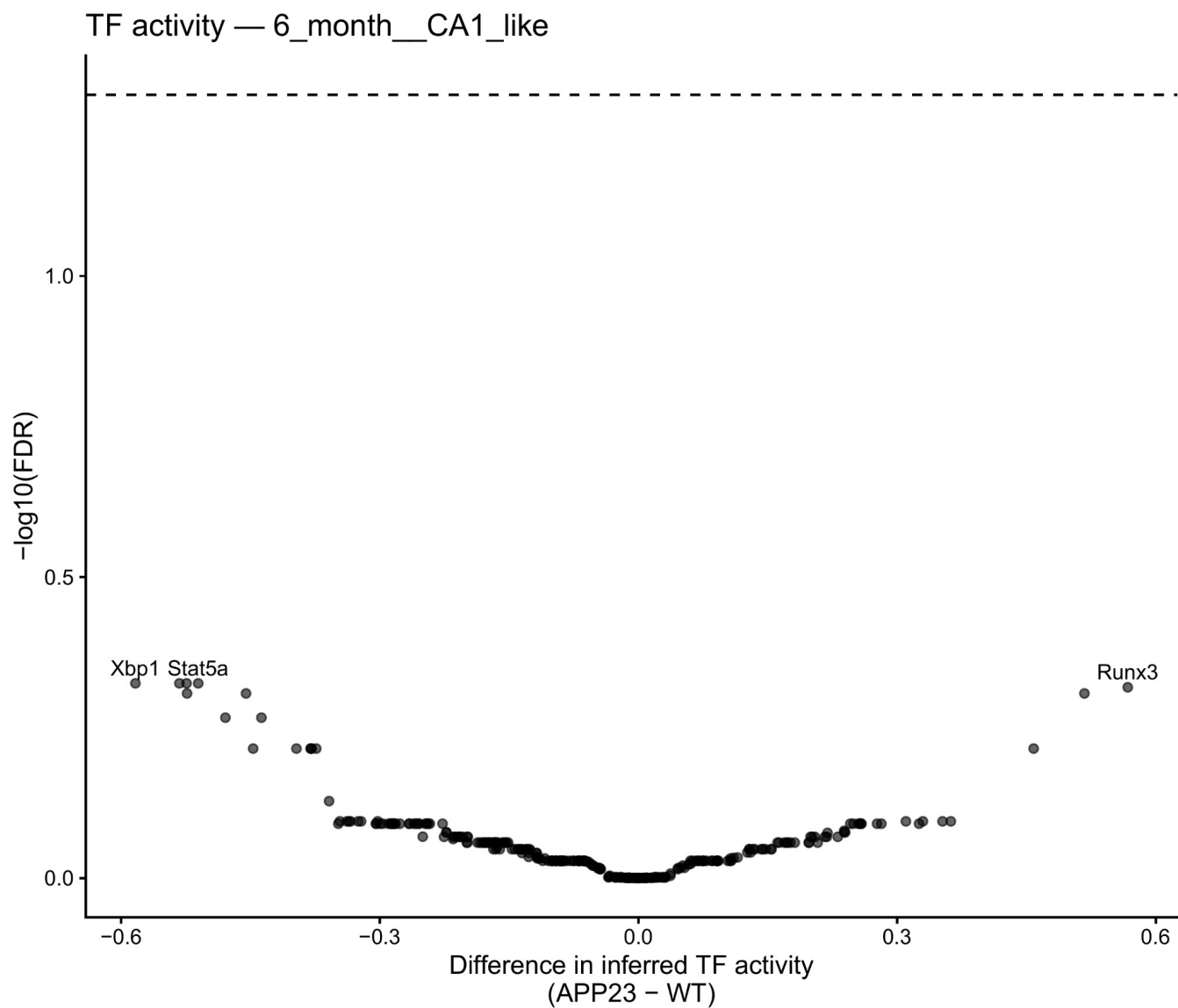
Supplementary Figure S8. Exploratory transcription-factor activity analysis.

A. 6-month DG.



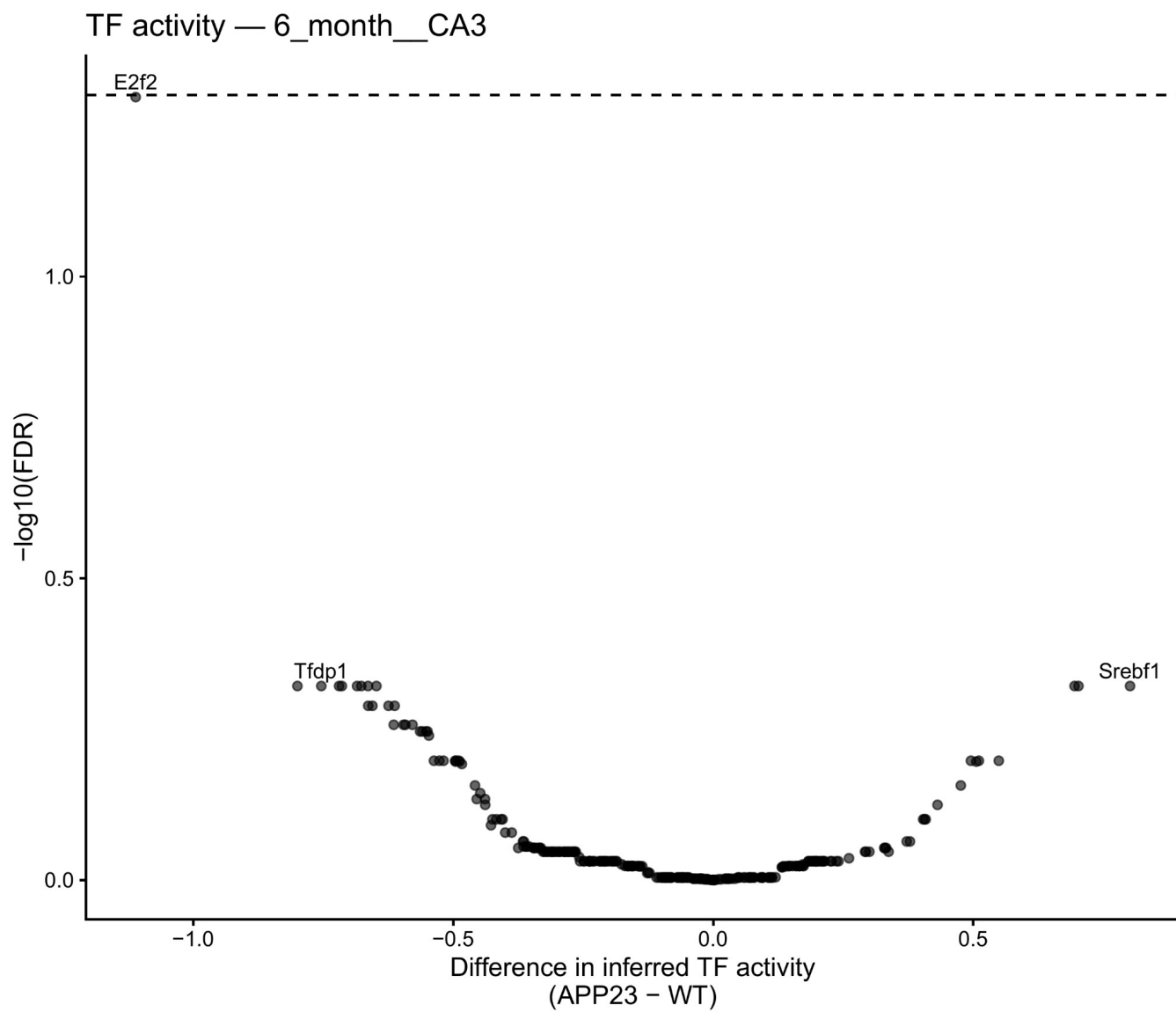
Supplementary Figure S8 (B)

B. 6-month CA1-like neurons.



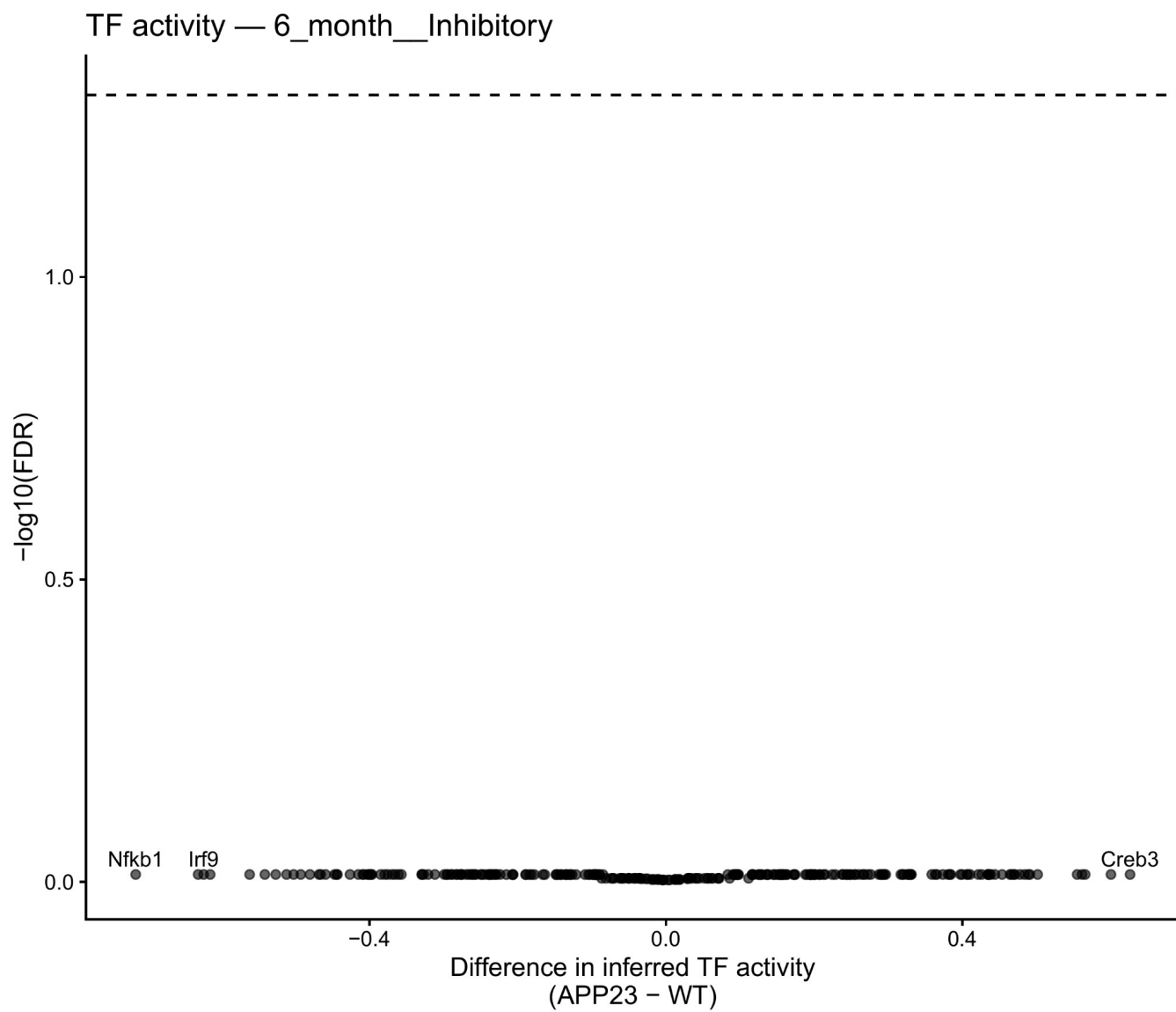
Supplementary Figure S8 (C)

C. 6-month CA3 neurons.



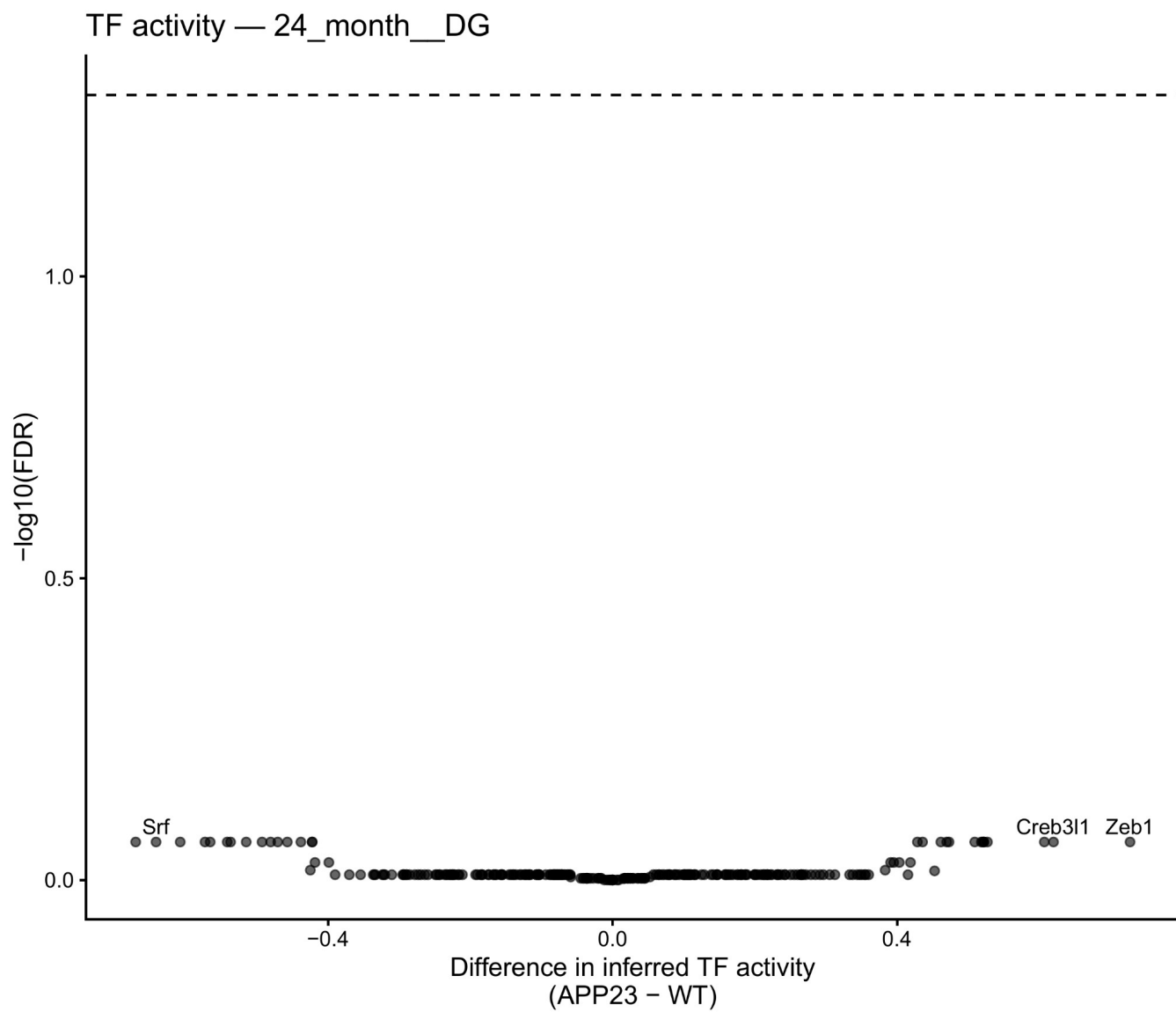
Supplementary Figure S8 (D)

D. 6-month inhibitory neurons.



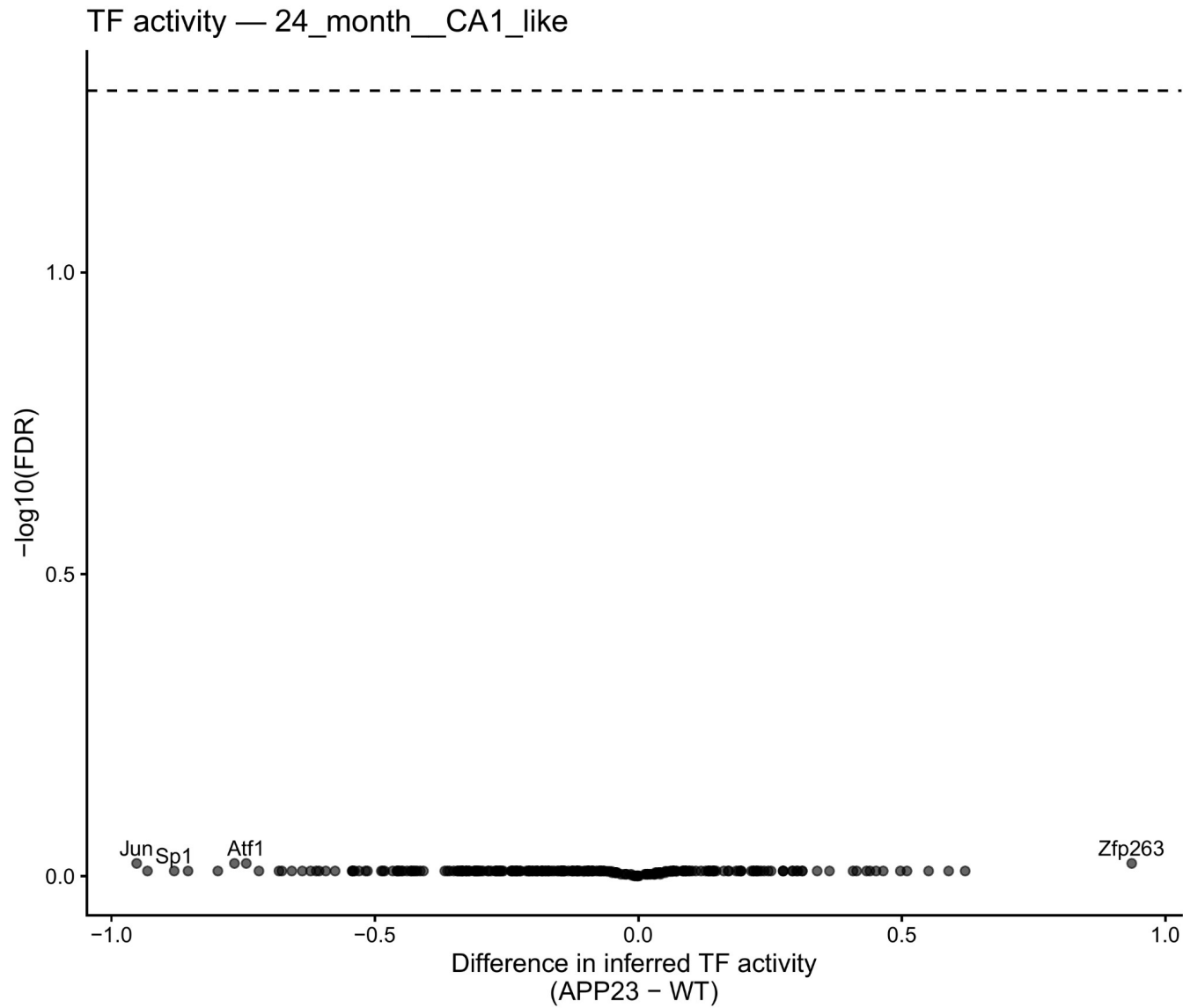
Supplementary Figure S8 (E)

E. 24-month DG.



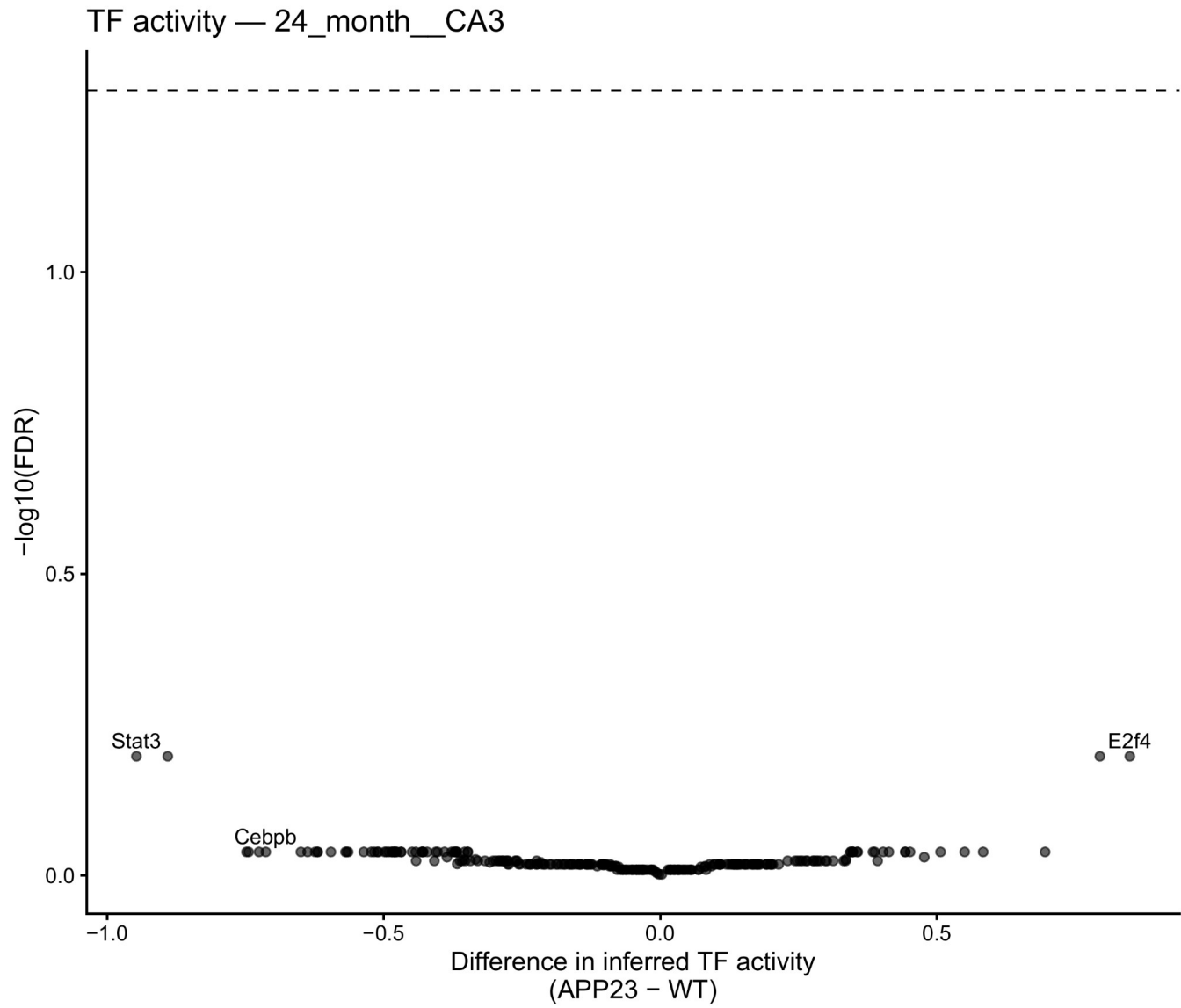
Supplementary Figure S8 (F)

F. 24-month CA1-like neurons.



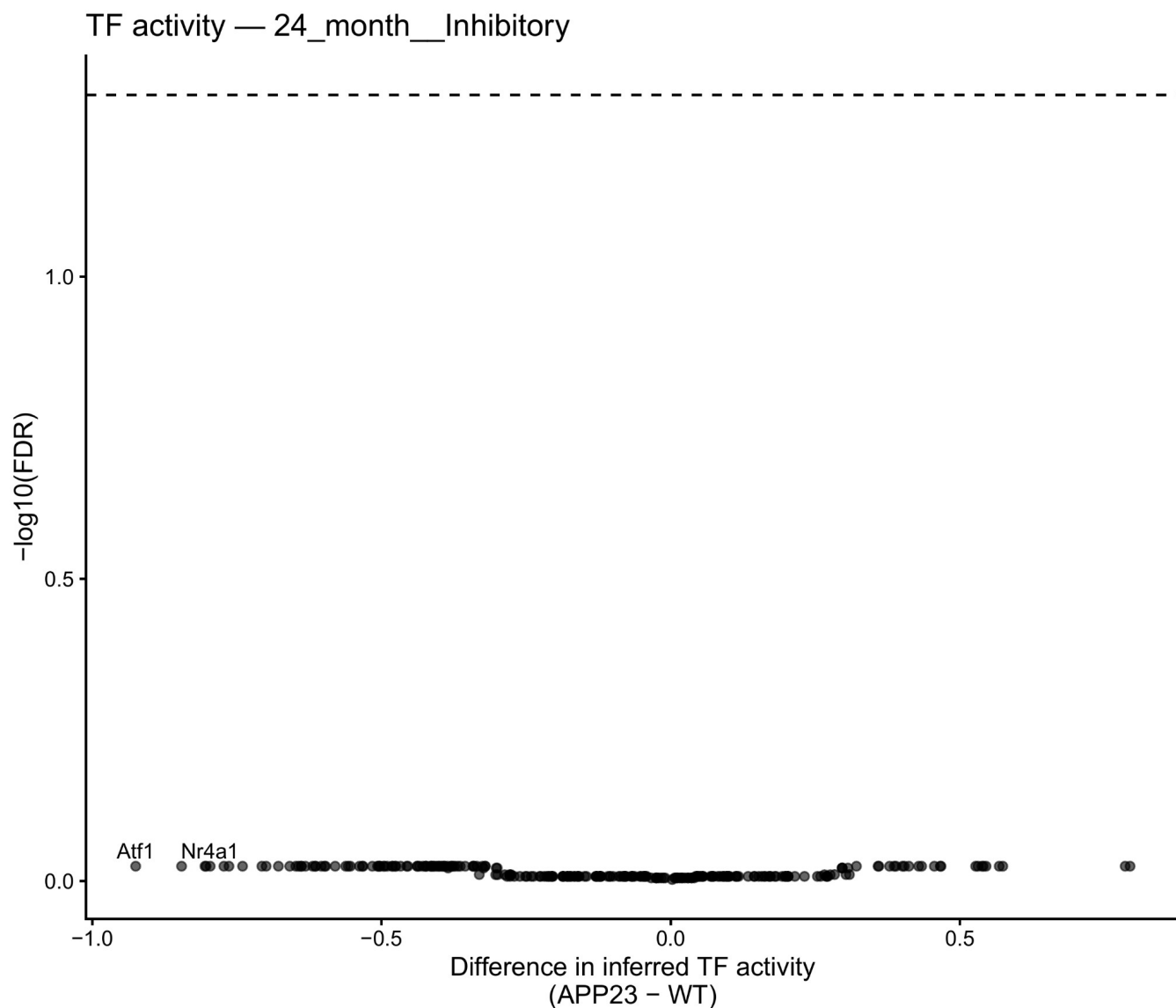
Supplementary Figure S8 (G)

G. 24-month CA3 neurons.



Supplementary Figure S8 (H)

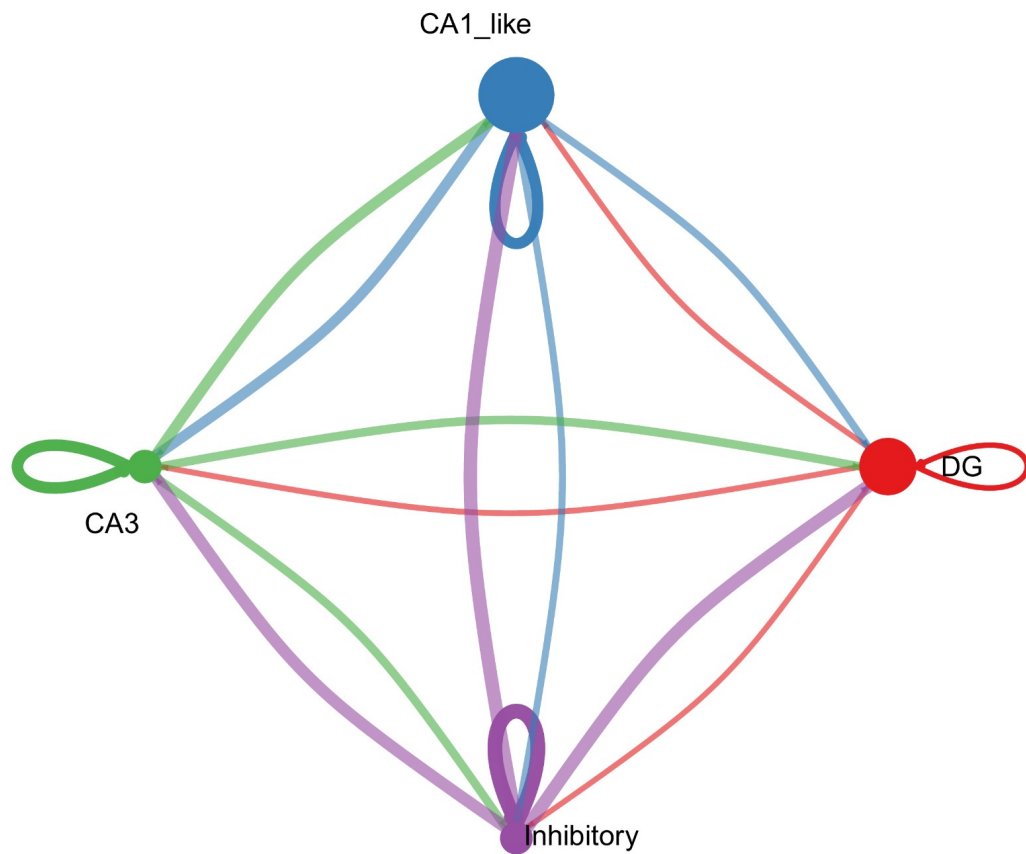
H. 24-month inhibitory neurons.



Legend. (A) 6-month DG. (B) 6-month CA1-like neurons. (C) 6-month CA3 neurons. (D) 6-month inhibitory neurons. (E) 24-month DG. (F) 24-month CA1-like neurons. (G) 24-month CA3 neurons. (H) 24-month inhibitory neurons. TF activity was inferred using DoRotheA/decoupleR and compared between APP23 and WT within age and neuronal subtype. The dashed line denotes FDR = 0.05; no TF activity comparison passed FDR < 0.05, so labeled regulators are exploratory.

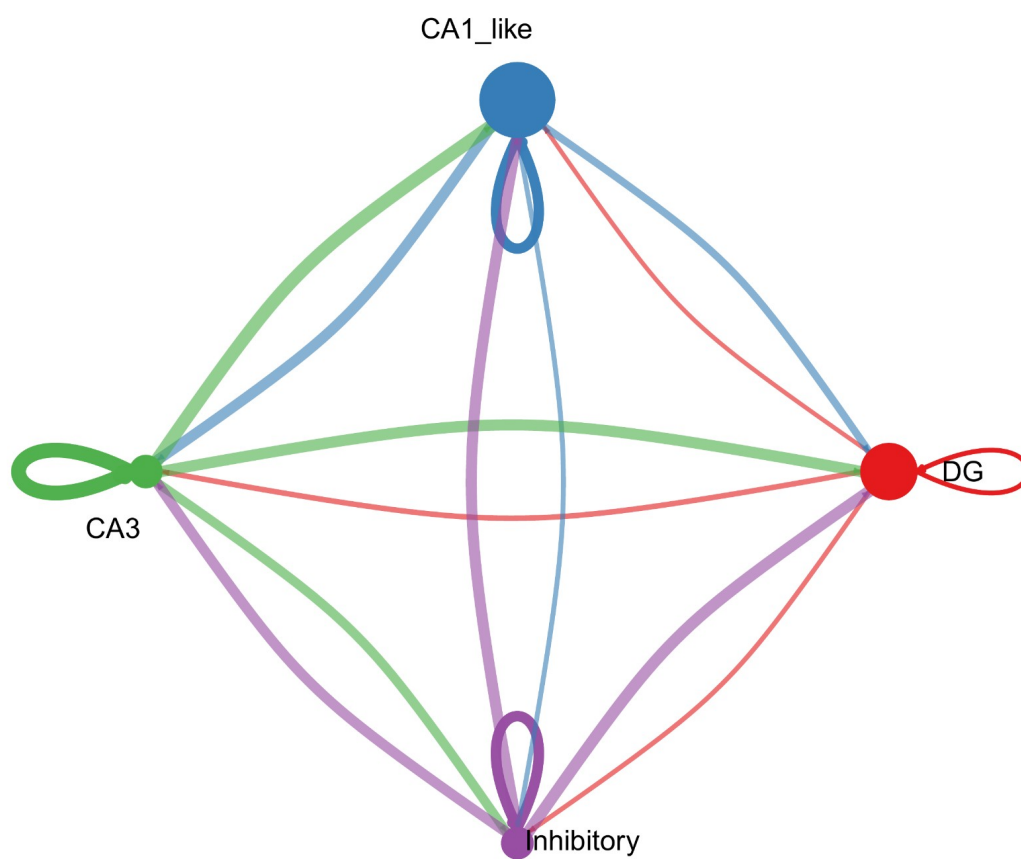
Supplementary Figure S9. CellChat communication networks within 24-month WT and APP23 hippocampal neurons.

A. WT interaction number.



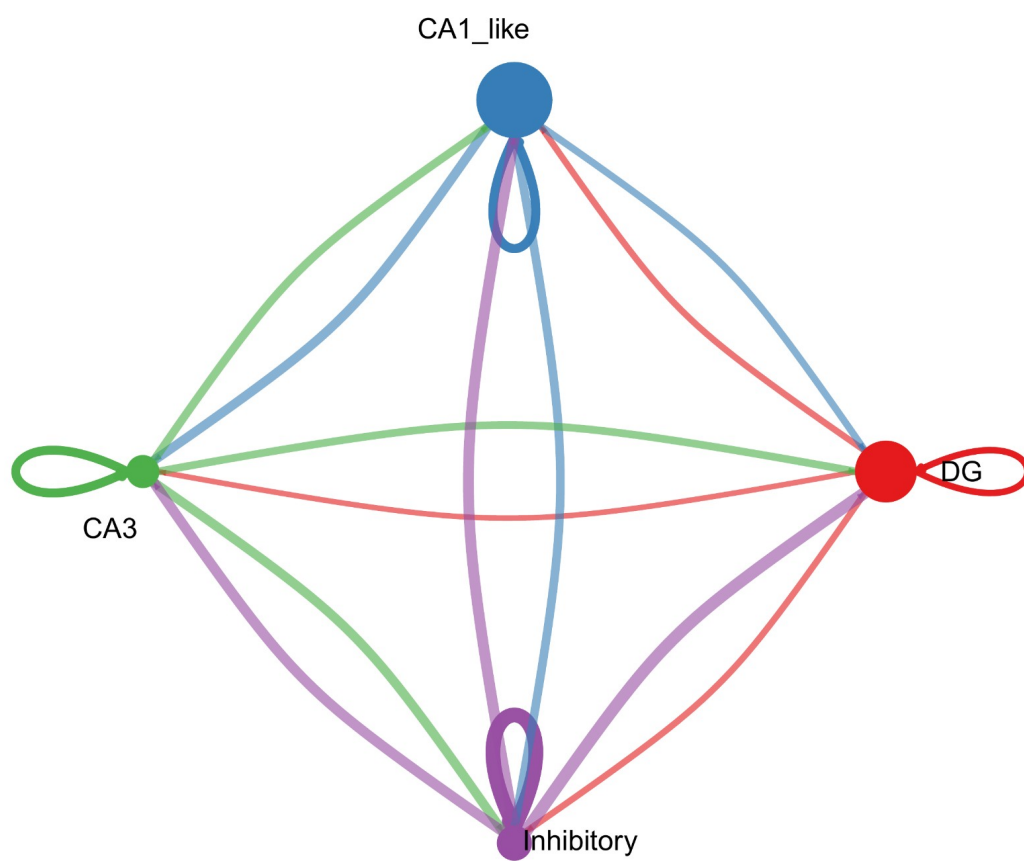
Supplementary Figure S9 (B)

B. WT interaction strength.



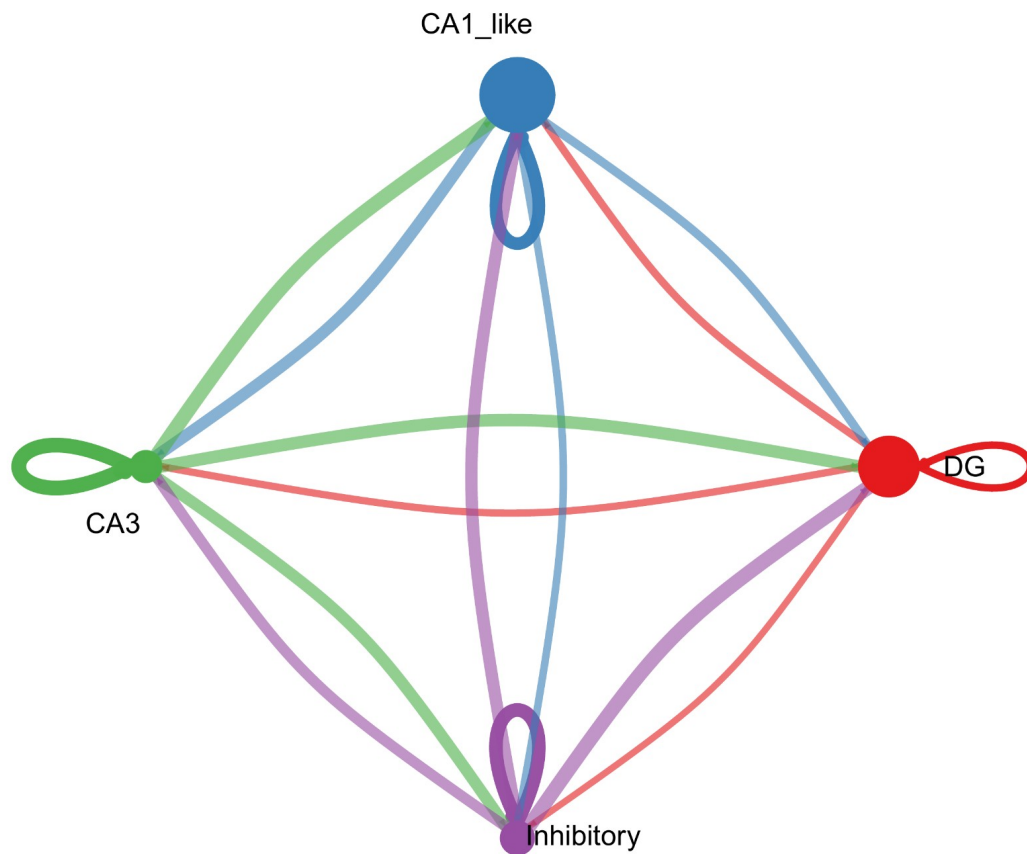
Supplementary Figure S9 (C)

C. APP23 interaction number.



Supplementary Figure S9 (D)

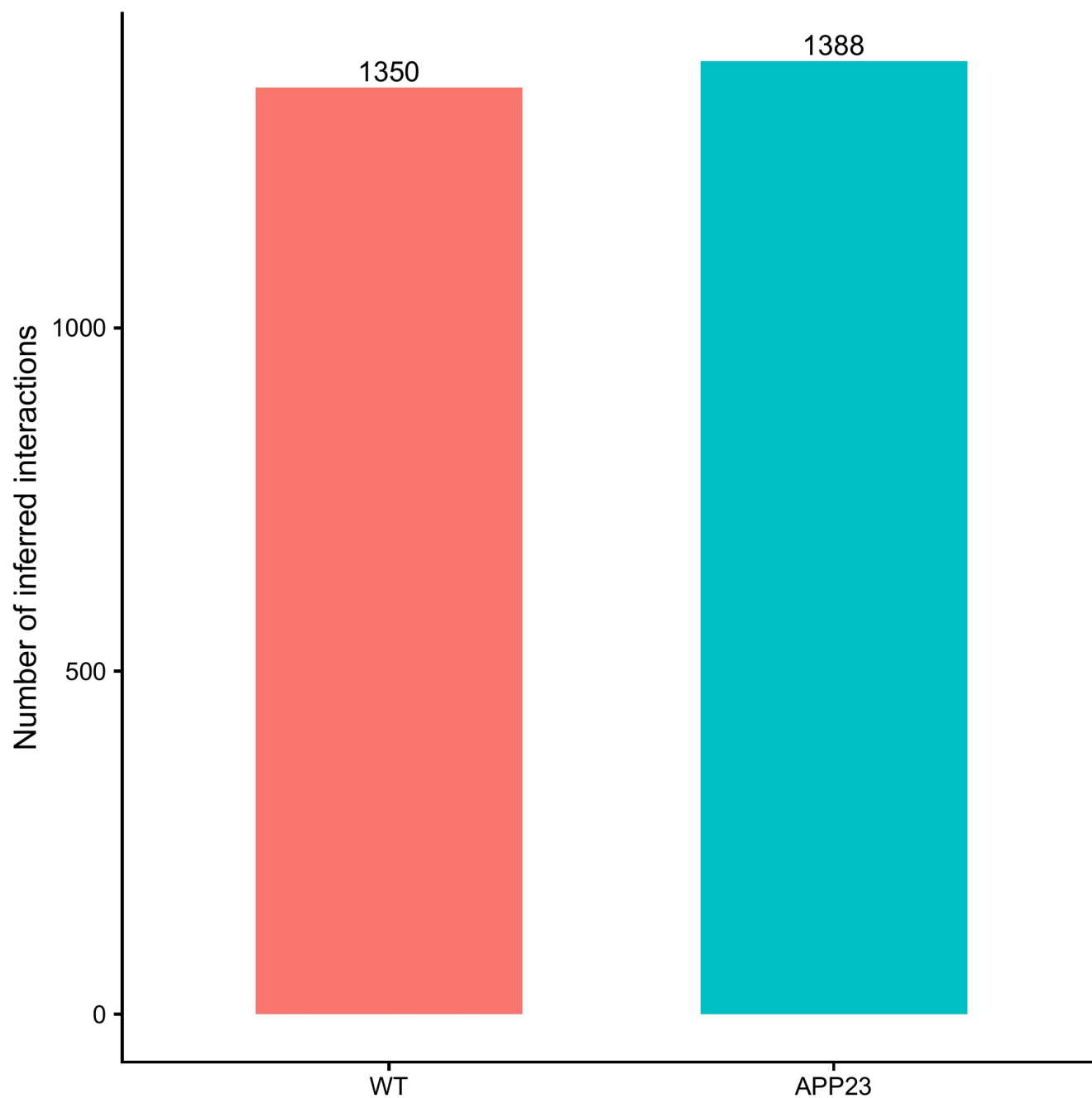
D. APP23 interaction strength.



Legend. (A) WT interaction number. (B) WT interaction strength. (C) APP23 interaction number. (D) APP23 interaction strength. CellChat results are descriptive condition-level inferences from neuronal nuclei and should not be interpreted as mouse-replicate-level causal evidence.

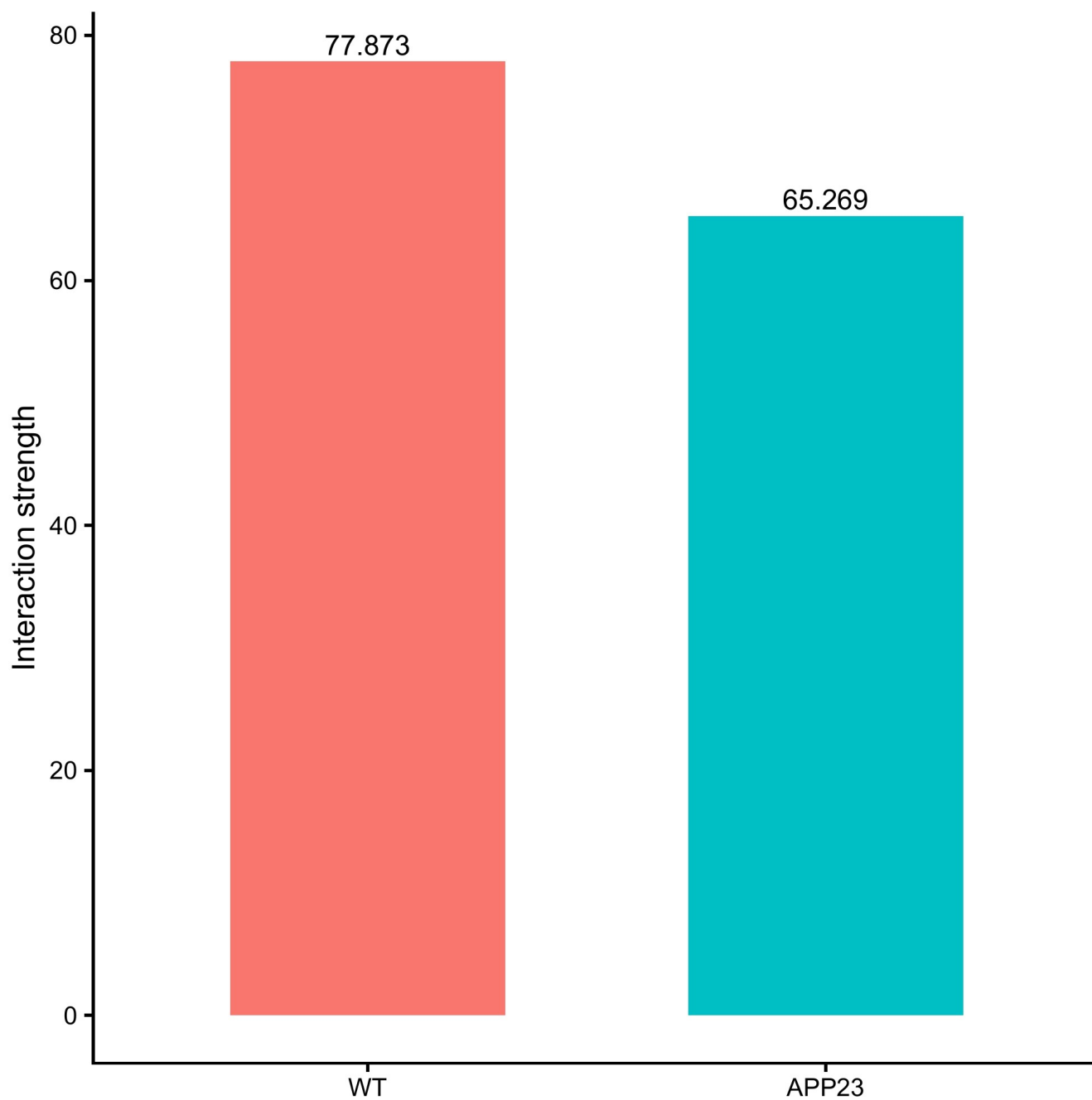
Supplementary Figure S10. CellChat global comparison of WT and APP23 neuronal communication.

A. Comparison of total inferred interaction number.



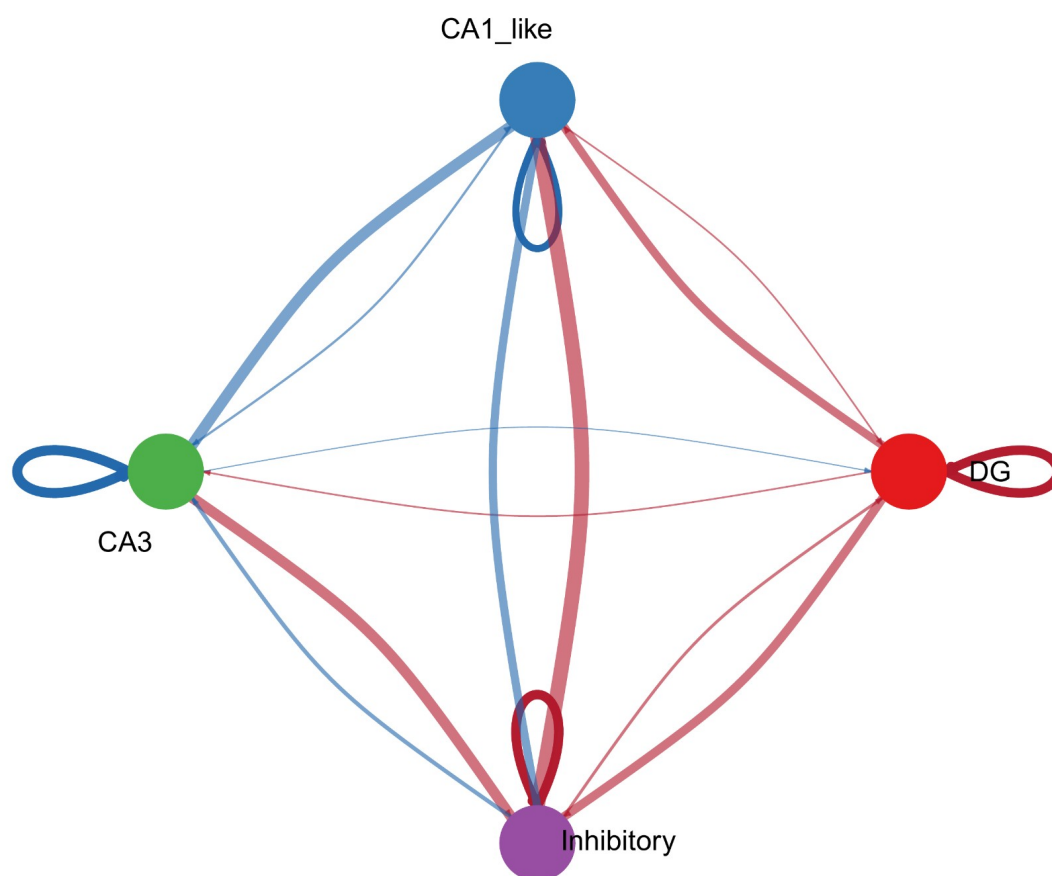
Supplementary Figure S10 (B)

B. Comparison of total inferred interaction strength.



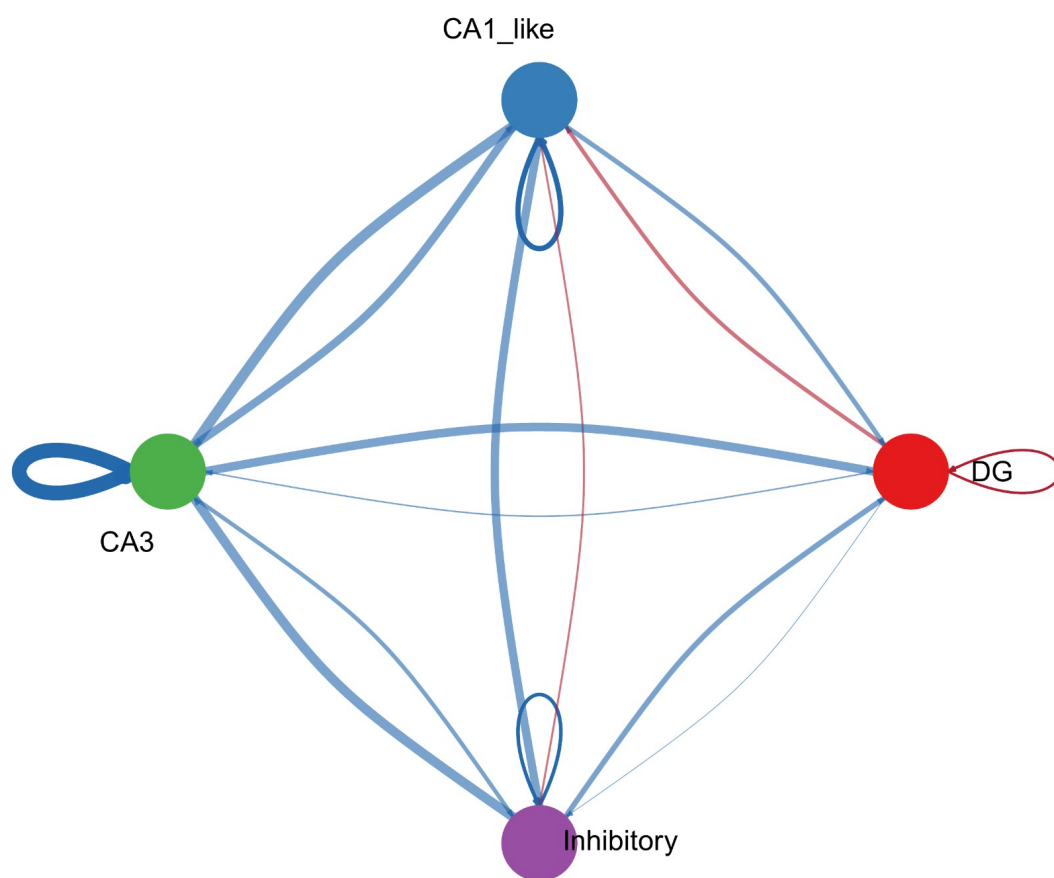
Supplementary Figure S10 (C)

C. Differential interaction number, APP23 minus WT.



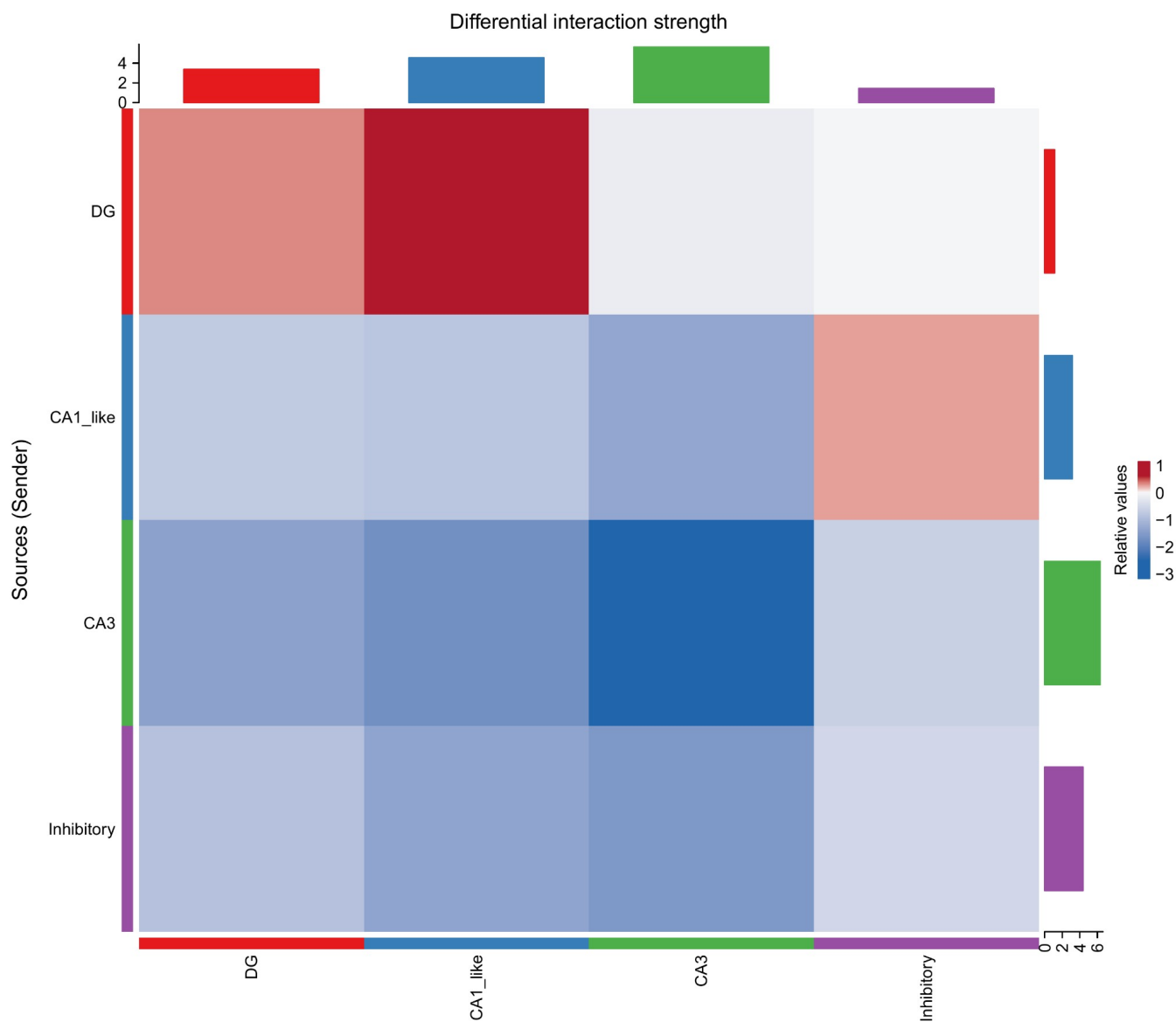
Supplementary Figure S10 (D)

D. Differential interaction strength, APP23 minus WT.



Supplementary Figure S10 (E)

E. Heatmap of differential interaction strength.



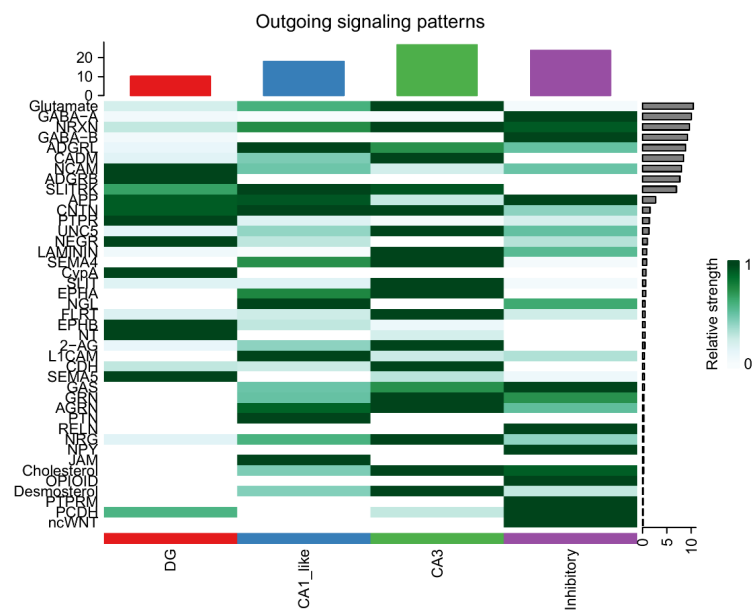
Legend. (A) Comparison of total inferred interaction number. (B) Comparison of total inferred interaction strength. (C) Differential interaction number, APP23 minus WT. (D) Differential interaction strength, APP23 minus WT. (E) Heatmap of differential interaction strength. CellChat results are descriptive condition-level inferences from neuronal nuclei and should not be interpreted as mouse-replicate-level causal evidence.

A. WT incoming signaling roles.



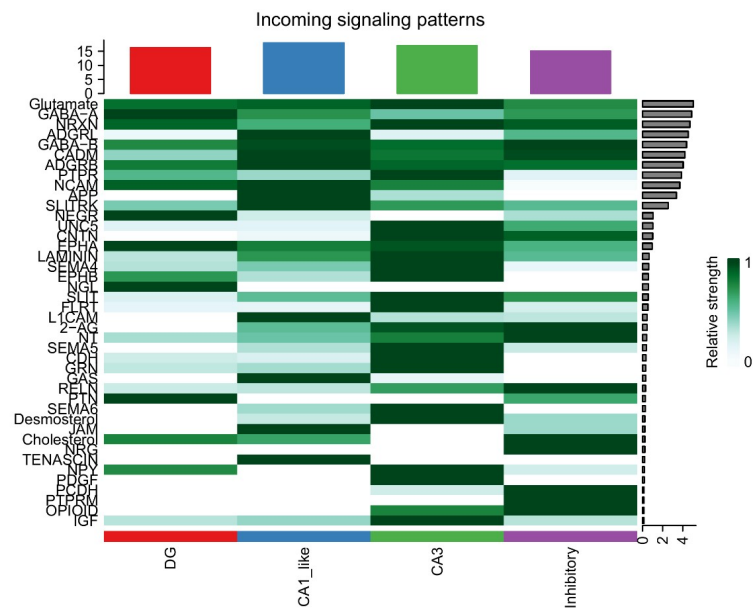
Supplementary Figure S11 (B)

B. WT outgoing signaling roles.



Supplementary Figure S11 (C)

C. APP23 incoming signaling roles.



D. APP23 outgoing signaling roles.

